

FH Aachen

Bachelor's Thesis

in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science

**AI Agents in Software Development: Best Practices for
Structured Human-Agent Collaboration**

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Declaration

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During the preparation of this thesis, I made use of AI-based tools to support literature research and to assist in rewording and refining the phrasing of certain passages. All underlying analysis, arguments, conclusions, and the overall structure of this thesis are my own. Any AI-assisted output was reviewed, verified, and edited by me before inclusion.

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1 Introduction

1.1 Background

Large Language Models have moved a long way from being simple autocomplete engines for code. Today, tools built on top of them act as genuine coding agents: they can read a set of requirements, plan an implementation, write entire features, generate their own tests, and reason, at least to some degree, about the architecture of the software they are building. Modern coding agents can be integrated directly into conventional development environments, allowing developers to hand off substantial parts of the implementation work to an AI agent while still reviewing and supervising what it produces.

More recently, agent-first development environments have emerged that extend this concept further. Rather than relying on a single agent working through a linear sequence of tasks, these environments can support the orchestration of several specialized agents, each responsible for different parts of the development workflow, from planning and architectural design to implementation and verification.

This shift changes the nature of software development in a subtle but important way. In traditional development, the person writing the code is also the one reasoning about it. When an AI agent takes over the implementation, a layer is inserted between the idea and the code: the agent interprets instructions, makes its own architectural choices, and produces artifacts that a human then has to review, correct, or approve. How good the resulting software turns out to be depends not only on how capable the underlying model is, but just as much on how the collaboration between human and agent is set up in the first place — how precise the prompts are, whether any architectural boundaries are defined at all, and what kind of review process, if any, the agent’s work has to pass.

1.2 Motivation

This thesis grew out of a practical project in which software was built from the ground up using AI coding agents, across two development phases that each embody a different philosophy of working with an agent.

The first phase followed a deliberately loose approach: an agent was asked to implement features from scratch with structural guidance that began minimally and was extended incrementally over the course of development as specific problems surfaced. The goal was to observe how an AI agent behaves when given substantial implementation freedom within a single-agent, conversational workflow — and where exactly that freedom starts to cause problems, whether in the form of inconsistency, architectural drift, or small but consequential implementation mistakes.

What came out of that first phase shaped the second one directly. Rather than assuming that better software simply requires a more capable model, the second phase focused on something different: the architectural rules, review steps, and development discipline imposed on the agents themselves. A more structured methodology was introduced for this purpose, built around role-specialized agents working together under a defined process rather than one agent working through a single, linear thread of tasks. The specific tools, agent roles, and artifacts used to realize this structure are introduced in Chapter 3.

The motivation behind this thesis is therefore twofold. First, it aims to document and compare, in concrete terms, what changes when moving from an unstructured, single-agent approach to a structured, multi-agent approach in AI-assisted software development. Second, it aims to distill from that comparison a set of best practices for human-agent collaboration that hold up beyond the specific tools used here, and that can inform how developers approach this kind of work more generally.

1.3 Problem Statement

AI coding agents are being increasingly adopted in both industry and academic settings. However, systematic understanding of how the structure of human-agent collaboration — rather than solely the capability of the underlying model — influences the quality, architectural soundness, and reliability of resulting software remains limited. Existing research on AI-assisted software development has often focused on model capabilities, benchmarks, or individual code-generation tasks, while broader aspects of the development process — including requirements interpretation, architectural planning, iterative implementation, testing, review, and documentation maintenance — require further investigation.

Several specific challenges motivate the work presented in this thesis:

1. **Limited structural comparison.** Existing work provides limited empirical comparison between unstructured, prompt-driven interaction with AI coding agents and

more structured, architecture-first development approaches, particularly when both approaches are examined within the same software project.

2. **Unclear impact of architectural constraints.** It remains unclear to what extent architectural rules, mandatory design reviews, and the separation of responsibilities across specialized agents can improve the quality and consistency of AI-generated software compared with a single agent operating with fewer structural constraints.
3. **Gaps in testing and validation.** AI coding agents can generate tests that execute successfully without necessarily providing sufficient validation of the intended behavior. This raises the question of how testing and verification should be structured when substantial parts of both implementation and test generation are delegated to AI agents.
4. **Documentation drift.** As AI agents modify software iteratively, maintaining consistency between the implementation and related specifications, such as OpenAPI documents, becomes an important maintainability challenge.
5. **Limited consolidated guidance.** Despite the growing adoption of AI coding agents, empirically grounded guidance on how developers should structure human-agent collaboration across prompting, architecture, review, testing, and documentation remains fragmented.

1.4 Objectives of the Thesis

The aim of this thesis is to systematically document, analyze, and compare two contrasting approaches to AI-agent-assisted software development — one unstructured and one structured — based on a practical project, and to derive from this comparison concrete, actionable best practices for human-agent collaboration in software engineering.

More specifically, this thesis sets out to:

1. **Document the unstructured approach (Part 1).** Describe how the software was developed using a single-agent workflow with minimal architectural guidance, and analyze the resulting architectural limitations, the influence of prompt precision, and the extent to which human supervision was required.
2. **Document the structured approach (Part 2).** Describe the design and implementation of a structured, multi-agent development methodology, including the AI Development Charter, parallel role-specialized agents, mandatory design reviews, the

three-tier testing strategy, and the automation of OpenAPI specifications to prevent documentation drift.

3. **Evaluate each approach.** Define evaluation criteria covering code quality, testing, architecture, and development process, and assess each development approach against them.
4. **Compare the two approaches.** Conduct a direct cross-approach comparison to determine how the structured methodology affected the limitations observed during the unstructured development phase.
5. **Derive best practices.** Bring the findings together into a set of best practices for human-agent collaboration.

Taken together, these objectives position the thesis as both an empirical case study of AI-agent-assisted software development and a practically oriented source of recommendations that developers and organizations can draw on when structuring their own collaboration with AI coding agents.

2 Foundations

2.1 Large Language Models in Software Engineering

Large Language Models (LLMs) are, at their core, autoregressive sequence generators: given a sequence of tokens as input, the model predicts a probability distribution over the next token and generates output by repeatedly sampling from this distribution. Nearly all current LLMs are built on the Transformer architecture, which underlies their ability to generate long, coherent sequences of text — including code [1].

When applied to source code, this generative mechanism is unchanged in principle: code is treated as a token sequence like any other. Just as a language model predicts the next word of a sentence from the words that came before it, a code-generating model predicts the next token of a program from the tokens that came before it — where a token may be a keyword, a variable name, a punctuation mark, or a fragment of one. Given the beginning of a function signature such as `def calculate_total(`, the model does not "know" what a function is in any formal sense; it predicts, token by token, that a parameter name is likely to follow, then a closing parenthesis, then a colon, then an indented block — because sequences with this shape occurred overwhelmingly often in the code it was trained on. The model's fluency with syntax, naming conventions, and common patterns is therefore a learned statistical regularity, not an explicit understanding of a programming language's grammar or semantics.

The practical capability of LLMs applied to code has developed considerably since general-purpose language models were first adapted to this domain. Early applications were largely limited to local code completion: a model would suggest the rest of a line, or perhaps the body of a short function, based only on the few lines immediately surrounding the cursor — closer to a smarter version of an IDE's autocomplete than to an independent programmer. A notable step beyond this was Codex, a GPT-based model that started as a general-purpose language model and was then further trained — fine-tuned — specifically on publicly available code from GitHub, which made it capable of writing Python code from natural-language docstrings alone. This meant the model could take nothing but a function signature and a description of the intended behavior, such as

```
python
```

```
def is_palindrome(s: str) -> bool:
    """Check if a string is a palindrome, ignoring
    case and non-alphanumeric characters."""
```

and generate a complete, working implementation beneath it — predicting, token by token, what code plausibly follows such a docstring, based on patterns learned from the enormous number of similar functions it had seen during training. On the HumanEval benchmark introduced alongside it, Codex solved 28.8% of hand-written programming problems from their descriptions alone when generating a single attempt. When the model was instead allowed to generate up to 100 candidate solutions per problem and only one needed to pass, its success rate rose to 70.2% [2] — an early illustration that a single generated output is often unreliable on its own. This shift — from finishing a line to generating a working implementation from a description of intended behavior — is what makes it possible, in principle, to prompt a model with a feature description and receive an entire piece of software back, rather than a suggestion for what to type next. It is precisely this capability that current coding agents build upon.

These structural properties belong to the model itself, considered independently of any system built around it. A "call" to an LLM — a single request that sends in some text and receives generated text back — has no access to a filesystem, cannot execute code, and does not persist anything once it finishes responding, unless additional tooling is deliberately built around it to provide these capabilities. The limitations described below apply to the model in this bare form:

- **Fixed context window.** A model can only condition its output on a bounded amount of text supplied in a single call — its context window. Imagine asking a developer to fix a bug in one file of a large application, but showing them only that one file and none of the others it depends on: they might produce a change that looks correct in isolation while silently breaking an assumption made somewhere else in the codebase, simply because they never saw it. An LLM faces the same problem whenever a project's relevant files, conventions, or history exceed what fits in a single call. This limits the context available to the model when making decisions, can prevent it from reasoning across an entire unit of code at once, and means the model simply cannot process code that does not fit within the window unless the relevant information is explicitly and selectively supplied as input. This limitation is well documented specifically for coding LLMs: the difficulty of maintaining context over long sequences of code is considered a significant technical limitation of LLMs in code generation, since the functionality and correctness of code often depends on

information that lies outside of what fits in a single window [3]. A single LLM call therefore cannot be expected to hold an entire codebase, its full change history, or an extensive set of project-specific conventions in mind at once.

- **No persistent memory across calls.** Each generation is, by default, stateless: the model does not retain anything from a previous call unless that information is explicitly re-supplied as part of the current prompt. Concretely, if a developer asks a model to implement a login feature, and in a later, separate call asks it to write tests for "the login feature I just described," the model has no memory of the earlier conversation to draw on — from its perspective, "the feature I just described" refers to nothing at all, unless the original description is pasted back in. Any continuity across a multi-step development task must therefore be actively maintained by whatever system is orchestrating the calls, not by the model itself.
- **No inherent mechanism for verification.** An LLM generates output that is statistically plausible given its training and the supplied context, but it has no built-in means of executing that output, checking it against a specification, or confirming that it behaves as intended. This is a specific instance of a broader, well-studied phenomenon in LLM research known as hallucination — the generation of content that is fluent and syntactically well-formed but factually inaccurate or unsupported by the available evidence [4]. Applied to code, this means a model can generate a function that reads as entirely plausible — correct indentation, sensible naming, no syntax errors — while still returning the wrong result for certain inputs, calling a method that doesn't exist, or silently failing to handle an edge case the requirement demanded. The model has no way of catching this itself, since doing so would require actually running the code and comparing its behavior against the specification — a step the model itself cannot perform. The same applies to generated tests: a test that is syntactically valid and executes without error is not automatically a test that meaningfully validates the intended behavior.

Taken together, these properties mark the boundary of what a bare LLM call can be expected to do: it provides a generative and reasoning capability over token sequences, including code, but has no native ability to interact with a filesystem, execute code, retain state across multiple turns, or verify the correctness of its own output. Any system that requires these capabilities — as agentic coding tools do — must supply them externally, layered on top of the model rather than as an inherent feature of it. This distinction motivates the notion of a coding agent, introduced in the following section.

2.2 AI Coding Agents and Agentic Workflows

A bare LLM call, as discussed above, cannot access a filesystem, execute code, retain memory across calls, or verify its own output. A coding agent is what results when a system is deliberately built around an LLM to supply exactly these missing capabilities. The LLM still provides the underlying reasoning and text-generation capability, but it is now embedded in a loop that lets it observe its environment, take concrete actions in it, and adjust its next step based on what happens — rather than producing a single block of text and stopping. A common way of describing this is in terms of three capabilities that, together, enable autonomous behavior: planning, memory, and tool use.

From single generation to an iterative loop. A widely used way of structuring this loop, introduced under the name ReAct [5], has the model cycle repeatedly through three types of output instead of producing a single final answer: a reasoning step describing its understanding of the problem and its plan, an action it takes to gather information or affect its environment, and an observation of the result of that action, which then feeds into the next reasoning step. Instead of a plain LLM call, which receives an input and produces one finished answer, an agent structured this way produces a reasoning step, then an action, then reads back an observation of what that action actually returned — and repeats this cycle, each round informed by the last, until it judges the task complete.

Applied to a coding task, this loop might look concretely as follows. Given the instruction **"add input validation to the registration form,"** an agent does not simply generate a finished file in one step:

1. Reasoning: the agent reasons that it first needs to see the existing form implementation before it can change it.
2. Action: it takes the action of reading that file.
3. Observation: it observes the file's actual content — the current fields, structure, and existing logic.
4. Reasoning: based on that observation, it reasons about where validation logic should be added.
5. Action: it takes the action of writing the change to the file.
6. Observation: it runs the project's test suite as a further action and observes whether the tests pass.
7. Reasoning: if a test fails, it reasons about why, and the cycle continues — reading further code, revising the change, re-running tests — until the observation confirms the task is complete.

Each individual step in this loop is still, underneath, a single LLM call — stateless, bounded by a context window, with no ability to verify anything on its own. What the agent adds is the surrounding structure that decides what to feed into the next call, based on what came back from the previous one, so that seven or more individual calls end up behaving like one continuous, self-correcting attempt at the task.

Tool use. The "actions" in this loop are only possible because the agent is given access to tools: concrete operations it can invoke, such as reading or writing a file, running a shell command, executing a test suite, or querying an external service. A representative example of teaching a model to use tools is Toolformer, a model trained to decide which external APIs to call, when to call them, what arguments to pass to them, and how to incorporate their results back into what it generates next [6]. A coding agent applies the same underlying idea to a developer's toolbox instead of a calculator or a search engine: instead of an API for looking up a fact, the tools are things like "open this file," "apply this change," or "run this command and report what happened." This is precisely what compensates for the limitations of a bare model call — it has no way to read a file it wasn't shown or to check whether code it wrote actually runs, but an agent with file-reading and code-execution tools can do both, by making additional, purpose-specific calls to the model in between.

Memory and persistent context. An LLM, left on its own, does not remember anything between separate calls unless that information is explicitly resupplied. A coding agent addresses this not by giving the model memory it does not have, but by having the surrounding system track relevant state — which files have been read or changed, what the original task was, what has already been tried — and re-inserting the necessary parts of that state into each new call. This tracked state is typically kept as structured text or files maintained by the agent's surrounding system, outside the model itself, and is only pulled back in when relevant to the next step. From the developer's point of view, the agent appears to "remember" earlier parts of a multi-step task; underneath, this is the agent's surrounding system doing the remembering on the model's behalf, one call at a time [7]. This approach has a natural limitation: everything the agent chooses to re-insert still has to fit within the model's fixed context window, so as a task grows, the surrounding system has to be selective about what state actually gets carried forward.

Coding agents specifically. Because software engineering tasks are inherently complex, modular, and often require coordinating multiple steps, they were among the first domains where this kind of agent architecture was applied in practice. A coding agent built this way can be handed a task description substantially larger than a single function — implement a feature, fix a reported bug, refactor a module — and is expected to work through it the way the loop above illustrates: inspecting the relevant parts of a codebase, making a change,

running the project's own tests or build tools to check the result, and revising the change if that check fails, without a human needing to manually restate context or re-supply files at every step.

Single-agent versus multi-agent workflows. So far, this section has described an agent as a single loop: one model, cycling through reasoning, action, and observation to work through a task. Not every coding agent is structured this way. Some development environments instead coordinate several such loops at once, each running with a different role, a different set of tools, or a different part of the task — for instance, one agent responsible for planning an implementation, and a separate one responsible for carrying it out. In this arrangement, the agents do not merely take turns; each is a full reasoning-action-observation loop in its own right, and the environment is responsible for passing relevant output from one agent to another rather than leaving a single model to manage every concern at once.

2.3 Multi-Agent Systems and Role Specialization

The previous section closed on a distinction: a coding agent can operate as a single loop working through an entire task alone, or several such loops can be coordinated together, each with a distinct role. This section unpacks that second case.

Why split a task across several agents. A single agent handling an entire software task end to end — understanding the requirement, planning an approach, writing the implementation, and judging whether the result is any good — has to carry every one of these concerns within the same loop and, ultimately, the same context window. A conversation focused on breaking down what a feature should do is a different kind of reasoning from a conversation focused on writing the actual code for it, and both are different again from critically checking whether that code is correct. Asking one agent to move fluidly between all of these within a single continuous context means each concern competes for the same limited space and the same undifferentiated set of instructions. Splitting the task across multiple agents, each responsible for one of these concerns, allows each agent's context, instructions, and available tools to be tailored narrowly to what that concern actually requires, rather than one generalist agent attempting all of them at once.

Role specialization as an organizing principle. In systems built this way, agents are not interchangeable copies of one another carrying out the same job in parallel; each is assigned a distinct role with its own responsibility. A pattern that recurs across the literature on this kind of system is a division into roughly three roles: a *planner*, responsible for breaking a high-level task down into smaller, well-defined subtasks; one or more *executors*, responsible for actually carrying out a given subtask, such as writing a specific piece of code; and

a *critic* or *reviewer*, responsible for checking a completed subtask before it is accepted. A representative example of this approach applied specifically to software development is MetaGPT, a framework that assigns agents distinct roles modeled on a real software team — such as product manager, architect, engineer, and QA — and has them proceed through the same sequence a human team would, rather than leaving the agents to work out a division of labor on their own; this structure allows agents with more specialized responsibilities to verify intermediate results and catch errors along the way [8].

A concrete illustration. Consider a task such as "add a password-reset feature to the application." In a role-specialized setup, a planner agent might first break this down into subtasks — for instance, an endpoint that generates a reset token, a database change to store it, an email that delivers a reset link, and a page that lets the user set a new password — and pass that breakdown on. An executor agent then takes one such subtask, for example the reset-token endpoint, and works through it much like the single-agent loop described in the previous section: reading relevant existing code, writing the change, and running it. Once that subtask is implemented, a critic agent examines the result — not by redoing the implementation, but by checking it against the subtask's requirement and against the codebase's existing conventions, flagging anything that looks incorrect, inconsistent, or incomplete. If the critic finds a problem, that subtask goes back to the executor with the critic's findings attached, rather than being accepted as finished.

Coordination through structured hand-off. It is worth being precise about what "multiple agents working together" actually means in this architecture, since it is easy to imagine something closer to several people talking in real time. In practice, coordination usually happens through structured hand-off of artifacts rather than open-ended conversation: the planner's output is a concrete plan, passed as input to an executor; the executor's output is a concrete piece of code, passed as input to a critic; the critic's output is a concrete piece of feedback, passed back as input to the executor if changes are needed. Each agent is still, on its own, a self-contained reasoning-action-observation loop of the kind described in the previous section — what changes is that the input to one agent's loop is deliberately constructed from another agent's output, rather than from the original human instruction alone.

Trade-offs. Coordinating several agents this way is not without cost. Because each agent's work becomes an input to the next agent's reasoning, an error introduced early in the pipeline — an incomplete plan, a misunderstood requirement — can silently propagate through every later stage rather than being caught immediately, particularly if a later agent has no way of seeing the original task and can only work from what the previous agent handed it. A recent empirical study of exactly this kind of planner-executor-critic pipeline found that when a solution passes sequentially through such a pipeline, errors introduced

at one stage can quietly carry through to the next, motivating close attention to which agent is actually responsible when a failure occurs [9]. Beyond error propagation, splitting a task across several agents also means more individual LLM calls are needed to complete the same piece of work, since each hand-off requires the next agent's relevant prior context to be explicitly re-supplied rather than carried forward in a single accumulating context [10], and the system as a whole depends on each agent correctly understanding what the previous agent handed it — a coordination requirement that a single agent working alone does not face.

Taken together, role-specialized multi-agent systems trade the simplicity of a single continuous loop for narrower, more focused reasoning at each stage, at the cost of additional coordination overhead and new ways for errors to enter and spread through the process.

2.4 Human-in-the-Loop and Human Oversight in AI-Assisted Development

As we already saw, splitting a task across several agents does not remove the risk of an early mistake propagating undetected through the rest of a pipeline — it only changes where in the process an error is likely to be introduced. This raises a question that applies just as much to a single agent working alone as it does to a coordinated multi-agent pipeline: at what point, and in what way, does a human actually enter this process?

Human-in-the-loop as a design choice, not an incidental fact. It is tempting to describe human oversight loosely, as "a person eventually looked at the result." But in the literature on human-AI collaboration, human-in-the-loop (HITL) refers to something more specific: a deliberate design decision about where, in an otherwise automated process, human judgment is required before the process is allowed to continue, and what form that judgment takes. Human-in-the-loop AI combines automated capability with human oversight, feedback, and decision-making at various stages of a system's operation, rather than treating human involvement as something bolted on only at the very end [11]. This means the relevant question for any AI-assisted workflow is not simply "is a human involved?" but "at which stage, and with what authority?"

Forms of oversight. Human involvement can take noticeably different forms, and these are worth distinguishing rather than lumping together as "review." At one end, an agent's output can be accepted automatically, with no human check at all. At the other end, a process can require explicit human approval before it is allowed to proceed to the next stage — the process pauses at a defined point and genuinely waits, rather than continuing

regardless. In between sits review after the fact: a human examines a finished output once it already exists, and can accept, correct, or discard it, but the work leading up to that point was not gated on their approval. The literature draws a related distinction between keeping a human directly in a decision cycle versus having a human monitor an otherwise autonomous process and step in only when something looks wrong. Applied to the multi-agent example already introduced — a planner breaking a task into subtasks, an executor implementing one, a critic checking it — a human could sit at any of several points: approving the plan before any code is written, reviewing each subtask only after the critic has already accepted it, or not reviewing anything until the entire feature is finished.

Why placement matters, not just presence. The distinction between these forms is not merely procedural. If a human's only review happens once every subtask is finished, an error introduced at the very first step — a plan built on a misunderstood requirement — has already shaped every subtask built on top of it by the time anyone catches it, and correcting it may mean discarding work that depended on the flawed plan. Reviewing at an earlier gate, such as approving the plan itself before any implementation begins, catches the same class of error while it is still cheap to correct. This mirrors a long-standing observation in traditional software engineering, where a mistake caught during requirements or design is generally far less costly to fix than the same mistake caught only after it has been implemented and built upon. The same logic carries over directly to AI-assisted development: where a human is placed in an agent's workflow determines not just whether an error can be caught, but how much of the surrounding work has already been built on top of it by the time it is.

What human judgment contributes that the agent cannot supply on its own. As already discussed, an agent has no way of verifying its output against anything beyond what was explicitly given to it — it can check that its code runs, or that it satisfies a stated instruction, but it has no independent way of judging whether that instruction actually captured what was intended. A human reviewer can catch a different class of problem: cases where an agent followed its instructions correctly, and its output is internally consistent and free of obvious errors, but still does not achieve what was actually meant, because the instruction itself was incomplete, ambiguous, or simply did not anticipate a particular situation. This is a failure mode no amount of agent-side verification can catch on its own, since the agent has nothing else to check its work against.

Taken together, human oversight in AI-assisted development is best understood as a variable that can be tuned — in its placement, its frequency, and the amount of authority it carries at each point — rather than as a single, fixed ingredient that is either present or absent.

2.5 Prompt Engineering

As we already saw, oversight determines when a human checks an agent's work. This section addresses something further upstream: what the agent is actually told to do in the first place, before any output exists to check at all.

Prompt engineering as a deliberate practice. An LLM's behavior is shaped entirely by what is given to it as input — there is no other channel through which a person can influence what it produces, short of retraining the model itself. Prompt engineering is the practice of deliberately designing that input — the instructions, framing, and surrounding context supplied to the model — to reliably steer its output toward what is actually wanted. This involves crafting task-specific instructions, known as prompts, to guide a model's output without altering its underlying parameters, allowing the same pre-trained model to be adapted to very different tasks purely through how it is asked [12]. It is worth being explicit that this is not the same as simply phrasing a request politely or clearly by ordinary conversational standards — a request can read as perfectly clear to a human and still leave room for an agent to settle on an interpretation the human did not intend.

System prompts versus task-specific instructions. In practice, an agent's input is typically built from two distinct layers. A system prompt sets persistent context that applies across an entire session — the agent's overall role, general rules it should follow, or background information about the project it is working in. A task-specific instruction is the individual request made within that session — "add input validation to the registration form," for instance. The two serve different purposes: a system prompt shapes how the agent behaves in general, while a task instruction specifies what it should do right now. A poorly designed system prompt can undermine an otherwise precise task instruction, since the agent is still reasoning within whatever general framing the system prompt established — and, conversely, even a well-designed system prompt cannot compensate for a task instruction that leaves the actual requirement genuinely unclear.

Prompt precision and ambiguity. An underspecified instruction does not cause an agent to visibly fail or ask for clarification in the way one might expect; more often, it causes the agent to quietly settle on one interpretation among several equally plausible ones, without flagging that a choice was made at all. Consider a task instruction such as "write a function that returns the maximum value from a list." This reads as unambiguous at first glance, but it leaves a concrete question open: what should happen if the list contains a mix of types — integers, floating-point numbers, and strings together? Should the function return the largest number regardless of the other elements, or the largest value once every element is compared as if it were text? A precise instruction removes this gap by stating the assumption explicitly — for instance, "assume the list contains only numeric values" —

while an imprecise one leaves the agent to decide silently. This is not a hypothetical concern: recent empirical work on LLM-based code generation found that models frequently do not respond to this kind of ambiguity by producing varied, inconsistent attempts — which would at least signal that something in the task was unclear — but instead consistently commit to a single interpretation, producing code that is internally coherent and looks correct, while still not matching what was actually intended. This failure mode, in which a model converges on one confident but potentially wrong reading of an ambiguous task rather than exposing the ambiguity, has been documented across multiple established code-generation benchmarks [13]. The practical consequence is that an underspecified prompt is unlikely to produce an agent that visibly struggles — it is more likely to produce an agent that appears entirely confident while quietly having answered a different question than the one that was meant. Listing 2.1 shows an excerpt of a precisely specified prompt from this thesis’s own practical project, introduced in Chapter 4, illustrating what stating a requirement’s edge cases explicitly looks like in practice.

```
Required behavior: After a successful login, the backend/
frontend should determine the user’s default workspace and
redirect the user to a route like:
/workspace/:workspaceId

For example:
/workspace/1
/workspace/2

The workspace ID in the URL should represent the active
workspace being displayed.

Frontend requirements:
- Replace the generic /main route with a workspace-specific
route.
- After login, redirect the user to their default workspace
route.
- When the route is /workspace/:workspaceId, load the data for
that workspace only.
- If the user switches workspace from the dropdown, update the
URL to the selected workspace ID.
- The header should still display the active workspace name.
- Do not break the existing workspace dropdown, collection
sidebar, or overview tab behavior.
```

Listing 2.1: Excerpt of a precisely specified prompt for the workspace-scoped routing feature, from Part 1 of this thesis’s practical project.

Context design. Context is a separate concern from the system prompt versus task instruction distinction above — it can be placed in either layer, and is about what needs

to be said, not where it is said. Precision addresses whether an instruction states the actual requirement clearly; context design addresses something adjacent to it — whether the agent has been given the surrounding information it needs to act on that instruction appropriately, even when the instruction itself is perfectly precise. An instruction such as "add a new API endpoint for retrieving a user's order history" can be entirely unambiguous about what the endpoint should do, while still leaving the agent without the information it would need to implement it consistently with the rest of the project — the existing naming conventions for endpoints, the authentication pattern already used elsewhere, the framework and libraries already in use, or the format other endpoints use for error responses. Given only the bare instruction, an agent has no way to know these things and will invent plausible-sounding defaults of its own; given the same instruction alongside a short excerpt of an existing, similar endpoint, the agent has a concrete pattern to follow instead of a gap to fill. Precision and context therefore address two different gaps: precision narrows what is being asked for, while context supplies what the agent would otherwise have to guess.

Taken together, prompt engineering determines the starting conditions an agent begins a task from — not whether its output will later be checked, but how much ambiguity and missing context that output is generated under in the first place.

2.6 Software Architecture in the Context of AI-Generated Code

Prompt engineering and context design, as discussed above, address what an agent is given before it acts on a single task. This section addresses a different kind of problem — one that does not show up within any single task, but only becomes visible once many separate tasks have accumulated over time.

Architectural conformance and drift. A codebase's architecture is not just its current structure, but the set of intended structural decisions a project is meant to follow — how components are organized, how they are meant to communicate with one another, what belongs where. Architectural conformance is the degree to which the actual, implemented code continues to follow these intended decisions. The opposite of conformance is not a single event but a gradual process: the literature distinguishes architecture erosion, caused by decisions that directly violate the architecture, from architecture drift, caused by decisions made without awareness that a governing architecture exists at all — the outcome in both cases is the same, a growing gap between how the system was meant to be structured and how it actually is [14].

Why this is a cumulative problem, not a single-instance one. A single missing piece of context, of the kind discussed earlier, causes one decision to be made without the information it needed. Architectural drift is a different scale of problem: it is what results when many such individually reasonable decisions, made across many separate tasks over time, each fail to align with a structural pattern established somewhere else. No single one of these decisions needs to be wrong on its own terms — each may correctly satisfy the specific instruction it was given — for the codebase as a whole to gradually lose its coherence.

A concrete illustration. Suppose a project's first feature introduces a particular way of formatting error responses from its API — a consistent shape including a status code, a message, and an error identifier. Months later, a different task asks an agent to add a new endpoint elsewhere in the project. If that task's instruction says only what the endpoint should do, and nothing about the project's existing error-handling convention, whether the agent discovers that convention depends on whether it ends up searching the existing codebase for it — something coding agents are capable of, whether prompted to or on their own initiative, but not something that happens reliably every time. Left unguided, it may reasonably invent its own, differently structured error response. Considered on its own, this new endpoint may be entirely correct — it does what it was asked to do. Considered against the rest of the project, it has quietly introduced a second, inconsistent way of doing something the codebase had already settled on. Repeated across many features and many separate tasks, this is how a codebase drifts away from its own established patterns without any single change being clearly at fault.

Why this is not unique to AI agents, but is not the same either. Architectural drift is not a problem invented by AI-assisted development — human developers drift from an established architecture over time as well, particularly on projects with many contributors or a long history. What differs is how familiarity with the architecture is maintained. A human developer working on a project over weeks or months accumulates an implicit sense of its existing patterns simply by being continuously immersed in it. An agent's grasp of those same patterns, by contrast, is only as good as what is deliberately reconstructed for it at the start of each task — LLMs applied to code generation frequently produce code in isolation, without an understanding of project-specific architectural patterns, and this contextual disconnect is a documented source of style and structural inconsistency across generated code [15]. Without that reconstruction, an agent does not retain the accumulated sense of "how things are done here" that a long-tenured human contributor builds up passively.

Architectural drift is, by its nature, not something a single well-crafted prompt can solve. Precision and context, as discussed earlier, improve the conditions under which one task is

carried out, but drift is a property that emerges across many tasks over time — preventing it requires some form of constraint or convention that persists across all of them, rather than being restated freshly, or not restated at all, with every new instruction.

2.7 Software Testing and Test Validity

Architectural drift, discussed above, concerns whether generated code stays structurally consistent with the rest of a project over time. This section addresses a related but distinct question, one that applies even to a single, well-isolated piece of code: when a test passes, how much does that actually tell us?

A passing test versus a valid test. A test passing means only one specific thing: the code, when run against that particular test, produced the outcome the test checked for. It says nothing about whether the test checked for the right thing in the first place. A test suite can run to completion, report every test as passing, and even claim high coverage of the codebase, while still failing to verify that the software does what it was actually supposed to do — because the tests themselves never asked the right question. This gap, between a test passing and a test validating the intended behavior, is the central concern of this section.

Why this gap is particularly relevant when tests are agent-generated. As established earlier, an LLM has no inherent mechanism for verifying its own output against anything beyond what it was given — it can produce plausible, syntactically correct output, but has no independent way of confirming that output is actually right. A generated test is not exempt from this limitation; it is itself just another piece of generated code, subject to the same constraint. An agent asked to "write tests for this feature" can produce a test suite that runs successfully and reports strong coverage numbers, while the tests themselves check only shallow properties of the code rather than its intended behavior. This is not a hypothetical risk: the literature has found that code coverage and mutation scores are not reliable indicators of a test suite's ability to detect a bug when the code being tested may already contain one, meaning a test suite can exercise a large portion of the code without being any good at catching cases where that code is wrong [16].

A concrete illustration. Consider an agent asked to write tests for a password-reset endpoint. A test that only checks that calling the endpoint returns a successful HTTP status code will pass regardless of whether a reset email was actually sent, whether the generated token is valid, or whether the token even reaches the right user — the endpoint could be broken in exactly the way that matters to the feature's purpose, and this test would still report success. The test is not incorrect in what it does; it is incomplete in what

it checks, and that incompleteness is invisible from the outside once the test suite reports green.

A related but distinct failure mode: tests that validate the implementation rather than the requirement. A separate risk arises when the same agent, or the same generation process, produces both the implementation and its tests. In this situation, a test can end up confirming that the code behaves the way the code happens to behave, rather than the way the original requirement intended. Empirical work on LLM-based test generation has documented this concretely: when a model is prompted with a buggy implementation, it can treat the faulty behavior as intended and generate a test that asserts the incorrect outcome, rather than the behavior the function was actually supposed to have — the resulting test still passes, and still contributes to coverage, while actively encoding the wrong behavior as if it were correct [17]. This is a different problem from incomplete coverage: the test is not merely checking too little, it is checking the implementation against itself rather than against the original, independent requirement it was meant to satisfy.

Taken together, these observations point to the same underlying principle: whether a test suite is any good cannot be judged from whether it passes, or even from how much of the code it exercises, but only by checking it against the original requirement the software was meant to fulfill — a check the test suite itself, by construction, cannot perform on its own behalf.

2.8 OpenAPI Specifications and Documentation Drift

Test validity, discussed above, concerns whether a piece of code's behavior matches what was actually intended. This section addresses a related concern one level removed from the code itself: whether that behavior stays accurately described, in an artifact other people and other systems rely on, once the code has changed.

What an OpenAPI specification is. An OpenAPI specification is a structured, machine-readable document that describes an HTTP API's capabilities — its endpoints, the inputs each one expects, and the responses it returns — allowing both humans and computer programs to understand what a service offers without needing to read its source code or inspect its network traffic directly [18]. Because the specification is machine-readable rather than free-form prose, it is not only read by developers as documentation; it is also consumed directly by tooling — generating client code that calls the API automatically, generating interactive documentation pages, or validating that incoming requests match what the API claims to accept.

Why a specification and its implementation can diverge at all. An OpenAPI specification is, in practice, a separate artifact from the code that implements the API it describes — typically a separate file, written and updated by hand or by a separate generation step, rather than something the programming language itself enforces consistency with. Nothing structurally prevents the two from disagreeing: a specification file has no built-in awareness that the code it describes has changed, and the code has no built-in awareness that a specification file describing it exists at all, unless something is explicitly built to connect the two.

Documentation drift. This divergence — between what a specification claims about a system and what the system actually does — is generally referred to as documentation drift, and its underlying cause is structural rather than accidental: code changes are continuously exercised by compilers, automated tests, and continuous integration pipelines, so problems introduced into the code tend to be caught quickly, while documentation changes are not subject to any of these automatic checks and are instead updated manually, opportunistically, and often incompletely. This is conceptually the same shape of problem as the architectural drift discussed earlier — a gradual gap opening between an intended, documented state and the actual one — except here the divergence is not within the code itself, but between the code and a separate document describing it.

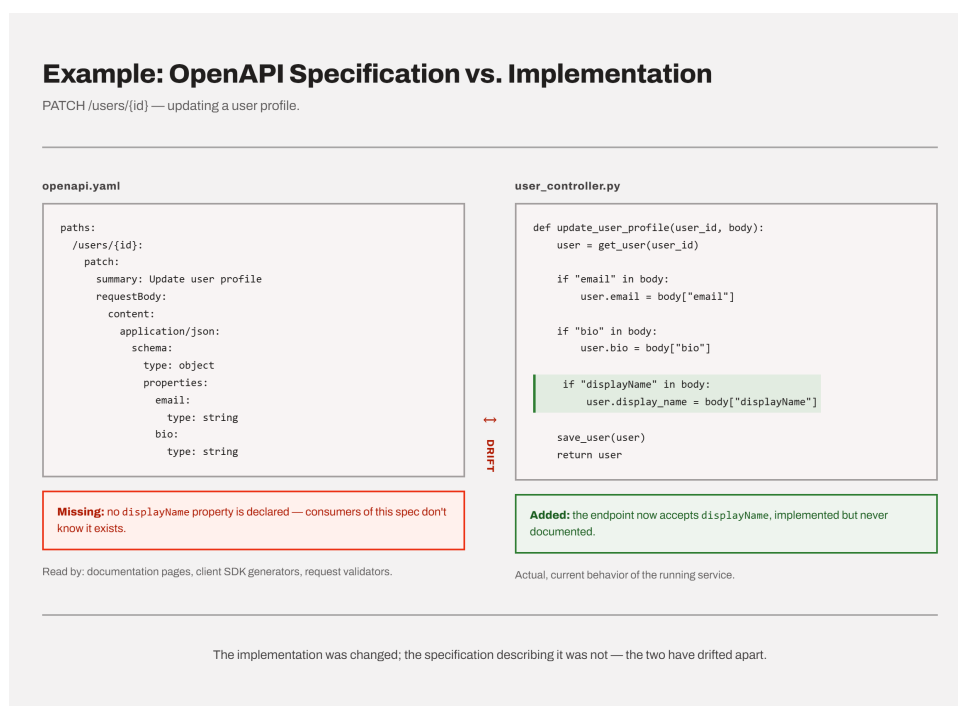


Figure 2.1: An implementation change accepted by the code but never reflected in the OpenAPI specification describing it.

A concrete illustration. Suppose an agent is instructed to add a new optional field to an existing endpoint's request body — for instance, allowing a user profile update to

optionally include a display name. Implementing this change means modifying the code that handles the request. Updating the OpenAPI specification to reflect that this new field now exists is a separate, additional step, and nothing about the instruction to "add the field" inherently requires the agent to also touch the specification file, unless it is explicitly told to or has been given a reason to check. If it is not, the implementation now accepts a field the specification does not mention — a consumer relying on the specification, whether a human developer or an automatically generated client, has no way of knowing the field exists at all, or may be shown an inaccurate picture of what the endpoint actually expects or requires.

Why this matters beyond documentation itself. With OpenAPI specifications specifically, a stale specification is not only a misleading document for a human reader — because these specifications frequently drive automated tooling directly, a specification that no longer matches the implementation can cause that tooling to actively misbehave: an auto-generated client built from the outdated specification may fail to send the new field at all, or a request-validation step may reject genuinely valid requests because the specification it is checking against has not been updated to allow them.

Like architectural drift, documentation drift is not something a single correctly specified instruction can prevent on its own — updating an implementation and updating its specification are two separate actions, and nothing automatically ties the two together across the many separate tasks that touch an API over the course of a project, unless something is deliberately built to keep them linked.

3 Methodology

3.1 Structure of the Practical Project (Two Sub-Projects)

This thesis is interested in how the structure of human-agent collaboration affects the quality of the resulting software — not in how capable one particular model is at isolated coding tasks. Standard coding benchmarks such as HumanEval, mentioned earlier in the foundations, or SWE-bench score an agent on short, self-contained problems with a single correct answer; they aren't built to capture what happens across a sustained, multi-feature project, where architecture, testing habits, and documentation either hold up or don't over time. For that reason, this thesis instead builds a single application under two different collaboration structures — an unstructured, single-agent workflow in Part 1, and a structured, multi-agent workflow with defined review steps in Part 2 — so that any differences in the resulting software can be attributed to how the collaboration was structured, rather than to a difference in what was being built.

The constant baseline: one software idea, one application. Both sub-projects implement the same underlying application, APICraft — a fullstack API collaboration and testing workspace that allows users to construct, execute, and manage HTTP requests, organize them into collections, inspect detailed response information, and persist requests for reuse. The software idea, its core modules, and its functional and non-functional requirements were written down once, at the very start of the practical project, and given to the agent only in Part 1. This initial specification document, together with early UI mockups, is what Part 1 was built from.

Part 2 did not receive this document again. By the time it began, the codebase already existed, and a persistent set of project knowledge files — describing the application's architecture and structure — carried the same context forward without needing to restate it. So while only Part 1 saw the original specification directly, both parts trace back to that same starting point: one software idea, developed continuously across the two parts.

Part 1: unstructured, single-agent development. The first sub-project was carried out using Claude Code inside Visual Studio Code, in a conversational, single-agent workflow,

with structural guidance limited largely to the software idea and requirements document, supplemented by a system-prompt file that itself began minimally and grew only in reaction to problems encountered along the way. Part 1 covers the application's foundational scope: authentication (login, registration, sign-out), workspace and collection management, request management, the request builder (parameter, header, body, and authorization tabs), request execution, response display, and the sidebar navigation.

Part 2: structured, multi-agent development. The second sub-project continues the same codebase produced in Part 1 — it is an extension, not a rebuild — but switches to Google Antigravity as an agent-first development environment and introduces an explicit collaboration structure: a three-agent workflow consisting of an Architect Agent, an Implementer Agent, and a Post-Implementation Review Agent, governed by a written set of architectural rules (an "AI Development Charter") that all three agents are required to follow. Under this structure, Part 2 both remedies architectural weaknesses identified during Part 1 and adds further scope to the application: workspace-level roles and permissions, inline request comments, an activity feed and audit logging, a performance and analytics dashboard, request export, and OpenAPI specification automation.

Artifacts produced by each part and their role in later analysis. Each part of the practical project produced its own evidentiary trail, which forms the primary source material for the comparison and evaluation that follow. Part 1 is documented through a review of its development process and the architectural problems that emerged during it, together with the reasoning applied when generated tests failed. Part 2 is documented through the structured artifacts its workflow requires by construction: an `implementation_plan.md` produced by the Architect Agent for each feature (including a mandatory design review of architecture rationale, state ownership, service ownership, testing strategy, and scalability concerns), a `walkthrough.md` and updated structure maps produced by the Implementer Agent, and a `review_report.md` produced by the Post-Implementation Review Agent against a fixed six-point checklist. Together with a persistent set of project knowledge files — an architectural rule set and living structure maps for the backend and frontend — these artifacts make Part 2's development process directly inspectable in a way Part 1's was not.

3.2 Tools and Models Used

Model. Both parts of the practical project were developed using the same underlying language model, Claude Sonnet 4.6. This was a deliberate choice: since this thesis is concerned with how the structure of a collaboration affects the outcome, holding the model

constant across both parts removes model capability as a confounding variable — any differences observed between Part 1 and Part 2 can be attributed to the collaboration structure and the presence or absence of process constraints, not to one part having access to a stronger or newer model. In Part 2, Claude Sonnet 4.6 was the model powering all three agents in the Antigravity workflow — the Architect Agent, the Implementer Agent, and the Post-Implementation Review Agent — rather than a different model being assigned to each role.

Development environments. Part 1 used Claude Code integrated into Visual Studio Code, and Part 2 used Google Antigravity — both described in Section 3.1.

Backend tooling. The backend was implemented in Python 3.13.2 within a virtual environment (with compatibility additionally checked against Python 3.12.4), using the Flask 3.1.3 web framework structured around an application-factory pattern. Backend testing was carried out with pytest 9.0.3, organized into 19 test modules covering the application's routes, services, and models.

Frontend tooling. The frontend was implemented in React 19.2.5, scaffolded with Create React App 5.0.1, and run on Node 20.14.0 with npm 10.7.0. Frontend testing used Jest together with React Testing Library for component-level smoke tests, consistent with the frontend testing conventions later enforced under Part 2's architectural rules.

API documentation tooling. The application's OpenAPI specification (`openapi.yaml`) was maintained by hand rather than generated automatically from code. To keep this specification synchronized with the actual API surface, a custom pytest-based drift-checker was written to compare the documented endpoints against the implemented routes, rather than relying on an external schema-validation library or code-generation tool.

3.3 Evaluation Criteria

Before Part 1 and Part 2 can be meaningfully compared, it is necessary to define, in advance, what "quality" means in the context of this practical project — otherwise any later comparison risks judging the two parts against standards invented after the fact, once their outcomes were already known. The criteria below come from the verification practices actually used while building both parts — they weren't invented afterward to fit the results. Each one is defined along with how it was assessed.

Specification satisfaction. The primary correctness criterion is not whether a feature's tests pass, but whether the feature satisfies the specification it was built against — its functional requirements, its non-functional requirements, and its edge cases. A feature

can pass a shallow test suite while still failing to handle malformed input, an unauthorized request, or an unusual but valid combination of parameters; specification satisfaction is judged by checking a feature's behavior against the original requirement, not only against whatever tests happen to accompany it.

Test validity. Because a test that runs without error is not automatically a test that validates the intended behavior, this thesis distinguishes between a test passing and a test being valid. This distinction emerged from a concrete case encountered during Part 1's development, discussed later, where a fix made a failing test pass without addressing the underlying problem it had correctly exposed. In Part 1 the lesson remained an informal, occasionally inconsistent practice; in Part 2, it was formalized into an explicit rule the agents were required to follow — before modifying a failing test, first explain why it failed, and determine whether the test, the implementation, or the underlying requirement is at fault before changing anything. Test validity is evaluated here by applying this same three-way diagnostic to both parts' test suites, regardless of whether it was formally enforced at the time.

Architectural conformance. Both parts are also evaluated against a set of concrete, checkable architectural rules, rather than a general or subjective notion of "clean code." These rules include limits on component size, a required separation between UI components and data-fetching logic, a defined ownership model for application state, and a prohibition on silently working around a failing test instead of diagnosing it. In Part 1, concerns of this kind were checked occasionally, as issues surfaced during development, but without a fixed, repeatable procedure behind them. Part 2 formalized the same underlying concerns into a six-point checklist, run consistently by its Post-Implementation Review Agent — covering violations of single-responsibility, unintended coupling between modules, untested code paths, excessive prop-drilling, unused or dead state, and inconsistent naming. Architectural conformance is evaluated here against this six-point checklist. Because Part 1 was not built with the full checklist in mind, its conformance is assessed only on the points its process actually checked for at the time; Part 2, built with the checklist already in place, is assessed against all six. This asymmetry — one part checked partially and informally, the other checked completely and by rule — is itself part of what the comparison in Chapter 7 shows.

Process and effort. Finally, both parts are considered in terms of the development process itself, not only its output: the number of iterations or corrections required to reach a working feature, the extent of human intervention needed to catch a problem the agent did not catch itself, and how much of the eventual solution had to be manually redirected rather than accepted as generated. This criterion draws on the milestone and task records kept throughout both parts, and is treated as complementary to the criteria above — a

feature can conform architecturally and satisfy its specification while still having required substantially more correction to get there, which is itself a meaningful difference between the two collaboration structures.

4 Part 1: Unstructured Development

4.1 Initial Situation and Objectives

Before development began, the practical project was framed around a small set of research points that Part 1 was designed to investigate directly: to what extent an AI agent can generate correct and usable code from a natural-language description, how reliably that code can be verified once generated, and how the way a task is formulated — the prompt, the context, the constraints given — affects the quality of the result. These questions guided every decision made during Part 1, from how the software idea was specified to how each feature was developed and tested.

The starting point was the software idea described in Section 3.1 — later expanded from a narrower initial "API tester" concept into the broader "API collaboration and testing workspace" ultimately built. This idea was translated into an explicit set of functional requirements (covering request building, execution, response handling, error handling, and persistence) and non-functional requirements (performance, security, reliability, robustness, usability, and maintainability). Success in Part 1 was defined not as "the generated code runs," but as specification satisfaction — code that fulfills the functional and non-functional requirements and correctly handles edge cases. This is a stricter standard than simply having tests pass: a feature could pass every test the agent wrote for it and still fail to meet the actual requirement, if the agent's own tests never covered that requirement in the first place.

Correspondingly, correctness in Part 1 was never assessed through a single signal. Verification combined execution-based testing (unit and integration tests, later extended toward property-based and fuzz testing), static analysis of the generated code, and human review of the agent's output — since execution-based testing only exercises the specific inputs and code paths its tests cover, leaving anything outside that scope unverified and any bug hiding there undetected, static analysis cannot verify runtime behavior and may produce false warnings, and human review, while able to judge things tests and static analysis cannot (readability, design quality, architectural decisions, correct interpretation of requirements), is time-consuming and subjective unless carried out under defined criteria.

4.2 Prompting Without a Structural Approach

Development in the first part had no predefined process for how prompts should be written or sequenced. The agent — used via Claude Code's chat sidebar in Visual Studio Code (Section 3.2) — was simply given a feature to build and prompted feature by feature, with the structure of each prompt worked out ad hoc rather than following a fixed template. The agent operated throughout under a persistent project file, `CLAUDE.md`, which supplied backend and frontend conventions and a set of architectural rules and is covered separately in Section 4.3. What this file did not provide was a per-feature planning or review step: no design was drafted or approved for a given feature before implementation began, and the file itself was updated reactively over the course of the project, as problems surfaced, rather than being fixed in advance and left unchanged. Listing 4.1 shows a representative example of this ad hoc structure, taken from the prompt that initiated the login component.

```
I created a react app in the frontend folder, i want to create
the first login components, i want you to develop this
components for me. I already created a Mockup for the
component and i want you to make it look exactly the same as
the mockup, so if you want to ask me something about the
mockup details it's okay. But don't change in the code until
you ask me.
@frontend/api-craft-app/src/components/login.js
```

Listing 4.1: Example prompt used to initiate development of the login component.

A further small set of rules was established early and carried through the rest of Part 1, layered on top of the project file and applied at the level of individual prompts: the agent was told to ask clarifying questions before generating code where the specification was unclear, and it was not permitted to modify any existing file without asking for permission first, so that every change could be reviewed before being applied. Later prompts added a further rule requiring the agent to briefly explain what it proposed to change and why before applying it. These rules shaped how the agent behaved within a session, but they did not amount to a planning step performed before a feature was started — there was no point at which a design was reviewed and approved ahead of writing code.

The typical pattern for a new frontend feature began with a mockup, sometimes accompanied by a short verbal description of details such as colors, fonts, and layout, followed by a single prompt asking the agent to reproduce it. The result was then compared against the mockup and refined through further prompts, usually several rounds of them, before the layout was considered close enough to the intended design. This was true even when

the mockup was provided upfront: matching it exactly in a single pass did not happen in practice, and the gap was closed through iterative correction afterward rather than through a more complete initial specification.

Backend prompts followed a different pattern, since there was no visual reference to match against. Instead of iterating toward a mockup, a single prompt typically specified the data model or endpoint involved, the validation rules to apply, and the expected behavior together in one request, often alongside explicit constraints on scope and performance. This combined pattern — data model or endpoint, validation, and constraints specified together in a single prompt — recurred across backend features throughout Part 1, in contrast to the frontend's iterative, comparison-driven refinement against a visual target. Listing 4.2 shows a representative example of this pattern, taken from the prompt that initiated the authentication feature.

```
I want you to implement a basic user authentication flow including both backend and minimal frontend integration. First, create a User model in the backend with the fields id (primary key), username (unique and required), and password (required), and store it in a database table. Then, implement a signup endpoint (POST /api/auth/signup) that accepts a username and password, validates that both are provided, checks whether the username already exists, and if not, creates and saves a new user in the database and returns a success response. Next, implement a login endpoint (POST /api/auth/login) that accepts username and password, validates them against the stored data, and returns a success response if correct or an error otherwise. On the frontend side, after a successful signup, the user should automatically be redirected to the login page, and after a successful login, the user should be redirected to a simple page displaying "Login successful." Keep the implementation simple without adding extra features such as JWT, sessions, roles, or OAuth, and do not introduce unnecessary complexity. Only implement the required logic and endpoints, and do not modify any existing files without asking for my permission.
```

Listing 4.2: Example prompt used to initiate development of the authentication feature.

Over the course of Part 1, individual task prompts grew progressively more detailed, but this growth happened reactively, in response to problems encountered in earlier features, rather than as part of a planned prompting strategy. Constraints that appear in later prompts — for instance, explicit instructions to avoid unnecessary re-renders, avoid repeated API calls, or not place API calls inside render logic — do not appear in the earliest prompts of the project. The same pattern holds for scope constraints such as "do not modify unrelated

files" or "do not add features beyond what is requested," which became a fixed part of later prompts after earlier, looser prompts had produced unwanted side effects. The prompting approach in Part 1 can therefore be described as incrementally, rather than deliberately, structured at the task level: the rules that accumulated in individual prompts came out of correcting specific problems as they appeared, not from a plan set before development began — and the same was true one level up, where the persistent project file supplying broader conventions started out with only a partial set of them and grew the rest the same reactive way.

4.3 System Prompt Context

Alongside the individual task prompts covered above, the agent operated under `CLAUDE.md`, the persistent project file introduced in Section 4.2, which set context applying across the whole development session rather than to any single feature. Where a task prompt specified what to build right now, this file specified how the agent was expected to behave and what conventions to follow throughout, and it stayed in place, read at the start of every session, for the whole of Part 1.

At the point development began, the file was limited to a small set of sections: an overview of the application, the technology stack, the project's folder structure, the core modules the application was meant to have, and a description of the agent's role and expected behavior. It also included a first set of backend and frontend guidelines, though only a partial one — general points such as using Flask Blueprints, JWT and Flask-Login for authentication, functional components and hooks on the frontend, and the Context API for global state were present from the start, without yet including the more specific architectural rules that would be added later. Listing 4.3 shows the Agent Role & Behavior section from the file as it existed at this early stage.

```
## Agent Role & Behavior
- You are the primary developer for both the backend (Flask)
and frontend (React) of this application
- Follow best practices at all times -- security, readability,
and maintainability take priority
- Explain your logic when implementing non-trivial decisions (
architecture choices, trade-offs, patterns used)
- Generated code must be modular, maintainable, and scalable
-- avoid tightly coupled or monolithic implementations
- Proactively suggest improvements if you identify a better
approach
- Always consider the full system (both frontend and backend)
when making decisions -- changes in one layer may affect the
other
```

Listing 4.3: Agent Role & Behavior section of CLAUDE.md, part of the file's initial content.

The rest of the file's content was added over the course of development, each addition following a problem encountered during implementation rather than being anticipated in advance. The clearest case is the React Architecture Rules section — the file naming convention, the component size limit, the requirement that all API calls go through a service layer, and the rule against reading application state such as user identity from more than one place. None of these were present when `MainPage.js` grew into a single component holding the top bar, sidebar, and main panel at once, when raw fetch calls were being made directly inside components, or when several components were independently reading user data from local storage. The CSS conventions — requiring every page to explicitly manage text selection and cursor behavior — were added the same way, after a problem involving the site's cursor behavior was encountered during development. The coding conventions, covering things like keeping business logic out of route handlers and not introducing speculative features, were likewise added only after related problems came up. The debugging scenarios the agent was expected to handle were added progressively rather than defined as a complete list from the outset. The "Known Design Decisions & Future Considerations" section could not have existed from the start either, since it documents specific issues found in the implementation, and by nature could only have been written once those issues had already been found.

The system prompt in Part 1 therefore followed the same pattern already observed at the level of individual prompts: it began with a limited set of instructions and grew incrementally, in reaction to specific problems as they occurred, rather than being defined as a complete set of architectural rules before development began.

4.4 Development of the First Feature (Login/Registration)

The login and registration feature was the first developed in Part 1, and it illustrates the same prompt design, generation, review, refinement, and testing cycle that recurred throughout the rest of development. As the module responsible for authenticating users and scoping the rest of the application to a single logged-in user, it was also treated as a foundation the later feature work would depend on.

The frontend was developed first. Development began with a visual mockup of the login page, given to the agent together with an instruction to reproduce it as closely as possible; the agent was told in advance to ask clarifying questions before generating any code, and additional details not fully captured by the mockup itself — specific colors, fonts, and layout choices — were provided before generation started rather than left for the agent to infer. Styling was explicitly constrained to Bootstrap, backend integration was deliberately postponed so the page could first be built as a UI-only component, and non-essential elements such as a "Forgot password" link were left as placeholders. The agent was also not permitted to modify any existing file without asking first. Listing 4.4 shows the additional details provided in response to the agent's clarifying questions, supplying exactly this kind of information before generation began (Figure 4.1).

```
1- Styling approach: for this project, use Bootstrap as the
primary styling approach. All components should leverage
Bootstrap's built-in classes for layout, responsiveness, and
general styling. No need for CSS Modules or styled-components
unless custom, non-Bootstrap styles are required.

2- Routing: Yes, we are using React Router for routing in this
project. The "Sign Up" link should navigate to the /signup
route, and the login route (or a similar authenticated route)
should navigate to the /MainPage route after successful
authentication. But know i want to start small with the login
page.

3- API integration: Now just the UI.

4- Forgot password: now just keep it a placeholder UI element.

5- Color details:
- Background: E9E3E3
```

Listing 4.4: Follow-up details provided in response to the agent's clarifying questions on the login page mockup.

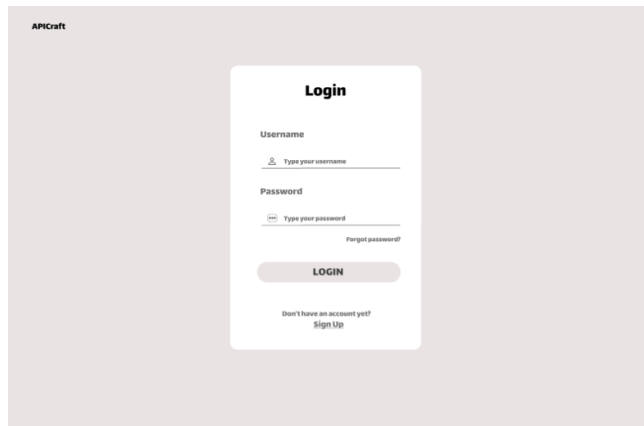


Figure 4.1: Mockup of the login page provided to the agent as the initial specification.

The first generated version matched the mockup closely in color and overall structure, but the login card was too narrow and sat too centered on the page, which did not match the intended design.

This was corrected through further prompts adjusting the card's width and the alignment of the input fields — an iterative process in which the first output served as a starting point rather than a finished result.

Across these iterations, the agent occasionally introduced CSS rules or code that were not needed for the page to function, which had to be identified and manually removed. The sign-up page was then built using the same mockup-first workflow and the same constraints.

The backend was developed afterward, in a single prompt (Listing 4.2) covering the data model, both endpoints, and the required behavior together. The prompt asked for a User model with an id, a username, and a password, a signup endpoint validating that both fields were provided and that the username was not already taken, and a login endpoint validating credentials against the stored data — along with minimal frontend integration to connect to the pages already built and the same requirement not to modify existing files without permission. This prompt did not ask for an email field at all; the omission was only identified afterward, once the specification was reviewed against what the feature was actually meant to support, and a follow-up prompt added it to the model and to both endpoints.

Reviewing the resulting implementation surfaced two further problems, neither of which had been anticipated in the original prompt. First, the database did not enforce email uniqueness, so multiple users could register with the same email address — a consequence of the original prompt never stating that the field should be constrained as unique. Second, the validation feedback returned to the user was too generic: both a missing username and

a missing password produced the same message, "Username and password are required," rather than identifying which field was actually missing, and no distinct message existed for the case of an email that was already registered (Figure 4.2).

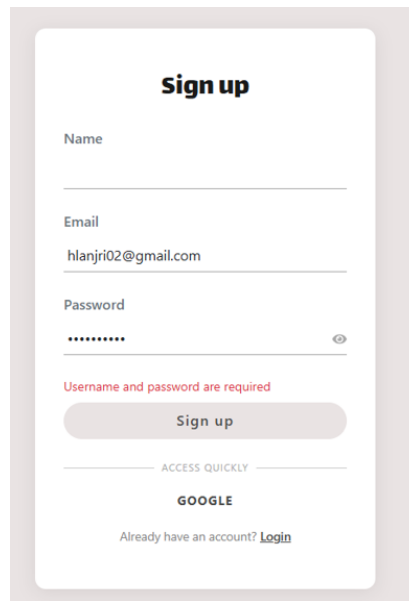


Figure 4.2: Sign-up form displaying the generic validation message that did not identify which field was missing.

Both were corrected through further prompts refining the backend validation logic and the corresponding frontend error handling, after which the messages returned distinguished a missing username, a missing password, and a duplicate email as separate cases.

Taken together, the missing email field, the missing uniqueness constraint, and the generic validation messages all trace back to the same cause: none of them were stated in the prompt that first specified the feature, and all three were only caught once the implementation was reviewed against what was actually intended rather than against what had literally been asked for.

4.5 Test Suites and Test Case Generation

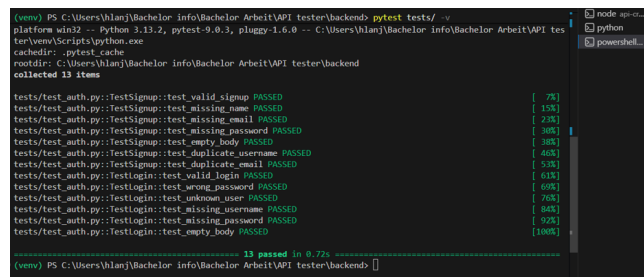
Testing of the login and registration feature covered only the backend.

Backend tests were generated only after the authentication logic had already been implemented and refined, using the prompt shown in Listing 4.5. The resulting suite covered valid signup requests, valid login requests, missing input fields, invalid credentials, duplicate usernames, duplicate email addresses, and empty request bodies, and was run against the backend authentication endpoints using `pytest`.

Based on the signup and login functionality, generate backend tests using pytest. Cover valid cases, invalid input, duplicate users, and wrong credentials. Do not modify existing code.

Listing 4.5: Prompt used to generate the backend test suite for the authentication feature.

The generated test suite itself served as a check on the refinements made during the earlier review: some of the generated tests would not have passed had the validation problems identified there not already been fixed. The test case for duplicate email validation, `test_duplicate_email`, is cited as an example — it checks for exactly the flaw the earlier refinement of the validation behavior had already corrected, meaning the test would have failed had that refinement not already been in place by the time the test was generated (Figure 4.3).



```
(venv) PS C:\Users\hlanj\Bachelor info\Bachelor Arbeit\API tester\backend> pytest tests/ -v
platform win32 -- Python 3.13.2, pytest 9.0.3, pluggy 1.6.0 -- C:\Users\hlanj\Bachelor info\Bachelor Arbeit\API tes
ter\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\hlanj\Bachelor info\Bachelor Arbeit\API tester\backend
collected 13 items

tests/test_auth.py::TestSignup::test_valid_signup PASSED [ 7%]
tests/test_auth.py::TestSignup::test_missing_name PASSED [ 15%]
tests/test_auth.py::TestSignup::test_missing_email PASSED [ 23%]
tests/test_auth.py::TestSignup::test_missing_password PASSED [ 30%]
tests/test_auth.py::TestSignup::test_empty_body PASSED [ 38%]
tests/test_auth.py::TestSignup::test_duplicate_username PASSED [ 46%]
tests/test_auth.py::TestSignup::test_duplicate_email PASSED [ 53%]
tests/test_auth.py::TestLogin::test_valid_login PASSED [ 61%]
tests/test_auth.py::TestLogin::test_wrong_password PASSED [ 69%]
tests/test_auth.py::TestLogin::test_unknown_user PASSED [ 76%]
tests/test_auth.py::TestLogin::test_missing_username PASSED [ 84%]
tests/test_auth.py::TestLogin::test_missing_password PASSED [ 92%]
tests/test_auth.py::TestLogin::test_empty_body PASSED [100%]

===== 13 passed in 0.72s =====
(venv) PS C:\Users\hlanj\Bachelor info\Bachelor Arbeit\API tester\backend>
```

Figure 4.3: Full authentication test suite passing after the validation refinements were applied.

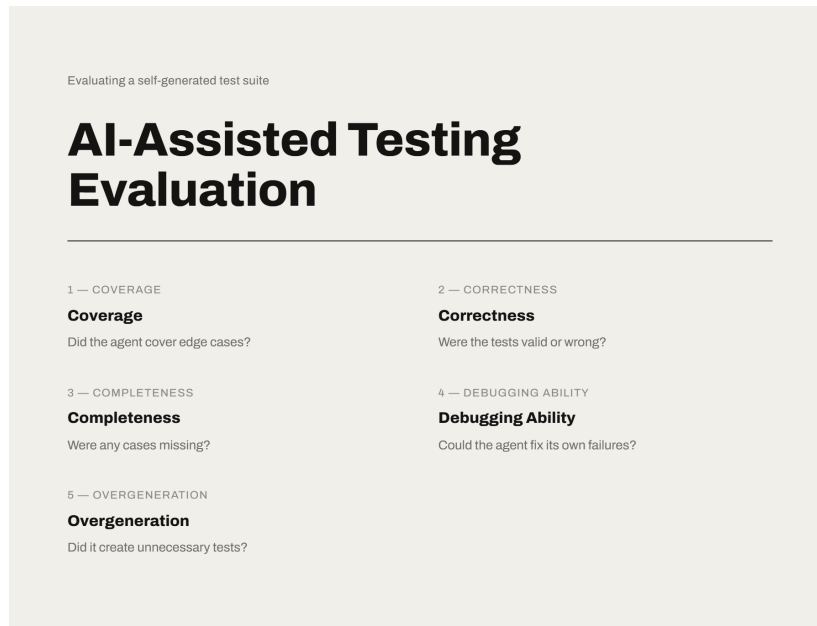


Figure 4.4: The five evaluation dimensions used in this thesis to assess the quality of the AI-generated test suite.

The generated suite itself was then evaluated along several dimensions (Figure 4.4). Coverage was judged broad, spanning the core authentication scenarios and their edge cases; correctness was judged sound, in that the tests reflected the intended backend behavior rather than only checking superficial conditions; and completeness was judged adequate, with no major authentication scenario identified as missing. An overgeneration check found no unnecessary or redundant tests — the suite stayed focused on the implemented authentication functionality rather than expanding into unrelated cases. A fifth dimension, the agent’s debugging capability — whether it could diagnose and fix a failing test on its own — was identified as a further evaluation criterion.

4.6 Architectural Analysis of the Generated Software

These same three problems — the missing service layer, the monolithic layout component, and the lack of centralized authentication state — were introduced earlier as the reason several of `CLAUDE.md`’s rules were added only partway through development. Here they are examined in full: what each problem actually was, why it occurred, and how it was resolved.

In each case, the problem was only identified after it had already spread across multiple files, and the fix took the form of a retrofit rather than a decision made in advance.

Listing 4.6 shows the React Architecture Rules section of `CLAUDE.md` that these three retrofits ultimately produced.

```
## React Architecture Rules (enforce on every new component)
- File naming: All component files must be PascalCase (e.g. `Sidebar.js`, not `sidebar.js`)
- Folder structure: Group components by feature under `src/components/<feature>/`. Shared/reusable pieces go in `src/components/shared/`. Auth pages go in `src/components/auth/`.
- Service layer: Every API call must live in `src/services/`. Components import and call service functions -- they never construct `fetch` calls themselves. The base URL lives only in `src/services/api.js`
- Component size: If a component exceeds ~100 lines or handles more than one visual region, split it. Pages are layout shells that own state; sub-components are presentational and receive props
- State ownership: State lives at the lowest parent that needs to share it. Do not duplicate state across siblings. Do not read `localStorage` in more than one place -- lift it into a context or the top-level parent
- No dead files: Delete placeholder, boilerplate, or abandoned components immediately -- do not leave them in the tree
```

Listing 4.6: React Architecture Rules section of `CLAUDE.md`, added in response to the three retrofits discussed in this section.

The first problem was the absence of a centralized service layer. Raw `fetch` calls were written directly inside React components — each component hardcoded its own base URL, defined its own headers, and handled errors independently, rather than going through a shared function. This had several consequences: changing the base URL required edits across every component that made a request, adding a global header such as an auth token had to be repeated in every call individually, the same logical API call could diverge between two components if each implemented it slightly differently, and components ended up responsible for both UI logic and data-fetching logic at once. The fix was to remove all raw `fetch` calls from components and move them into dedicated service files, with a single base URL defined once in a shared `api.js` module; components were changed to call service functions exclusively, without constructing HTTP requests themselves.

The second problem was excessive responsibility concentrated in a single component, `MainPage.js`. At 186 lines, it already owned three distinct regions at once — the top bar (workspace header, dropdown, tab strip), the sidebar (toolbar, search, collections navigation, drag handle), and the main panel (tab bar and panel body) — before the remaining planned features (collections, the request builder, the response panel, history)

had even been implemented. Left unaddressed, the file was projected to grow past 500 lines with dozens of tangled state variables, to the point that finding a bug in the sidebar would mean scrolling through request-builder code, and adding a new tab would mean touching the same file as the resize logic. The fix was to split the component into three focused parts: `MainPage.js` became a lean layout shell holding shared state and wiring the resize logic, while `TopBar.js`, `Sidebar.js`, and `MainPanel.js` each took ownership of one of the three regions, receiving what they needed through props rather than managing it themselves.

The third problem was the absence of a centralized authentication state. User identity — user ID, email, first name — was stored in local storage and read directly by whichever component needed it, independently of every other component doing the same: `MainPage.js` read the first name to display the workspace name, `WorkspaceDropdown.js` read both the user ID and the email on its own, and `TopBar.js` depended on workspace state that itself depended on the user ID. This produced three related consequences: there was no single place in the component tree representing who was logged in, so checking authentication meant validating local storage separately in every component that needed it; runtime changes such as logging out or changing the email had no way to propagate, since doing so would have meant finding and updating every individual read of local storage; and the problem would only get worse as more features needed user data, since each new component would be likely to repeat the same pattern rather than break from it. The fix was to introduce a single React context, `AuthContext`, that reads local storage once on mount and exposes the user object through a `useAuth()` hook; components stopped reading local storage directly and called the hook instead, and logging out was handled in one place, clearing storage and updating state together so that every dependent component re-rendered automatically.

None of these three problems were caught before they occurred, and in each case the individual decision that caused them was locally reasonable for the agent at the time it was made — the same pattern of architectural drift described earlier, playing out at the level of a single project instead of across separate tasks. What was missing was not correctness but a persistent point of reference: an existing service layer to extend, an existing component boundary to respect, an existing context to read from. Without one, each new piece of code was judged only against whether the feature worked, not against whether it stayed consistent with how the rest of the application already did the same thing.

4.7 Observations: Prompt Precision, Human Oversight, Architectural Limitations

Across the features covered so far, three observations recur often enough to be treated as general findings about Part 1 rather than as properties of any single feature.

The first is that the quality of the generated output tracked the precision of the prompt closely and directly. The frontend illustrates this at the level of visual detail: the closer a prompt specified what was wanted — layout, spacing, specific colors and fonts — the closer the generated result matched the intended design, while looser prompts left more for the agent to guess and produced results that needed further rounds of correction. The backend illustrates the same point at the level of missing requirements rather than visual detail: the missing email field, the missing uniqueness constraint, and the generic validation messages were not failures of the agent’s reasoning so much as direct consequences of a prompt that had never specified any of the three. In both cases, the gap between what was generated and what was actually wanted traces back to what the prompt did or did not say, not to a limitation in what the agent was capable of producing when asked precisely.

The second is that human oversight was present but shallow for most of Part 1, and this is what allowed the architectural problems discussed earlier to accumulate rather than being caught early. A standing rule prevented the agent from modifying any existing file without asking first, but in practice this meant changes were generally accepted once they worked, without being checked against the system they were being added to — the absence of a service layer, the growing `MainPage.js`, and the scattered local storage reads were all accepted this way, repeatedly, before any of them were treated as a problem. Oversight only became substantive once an issue was obvious enough to notice on its own — the file had already grown past 180 lines, or local storage was already being read independently in three separate places — at which point the fix required undoing an accumulation rather than preventing one decision. Had the same level of scrutiny been applied earlier and consistently, at the point each individual decision was made rather than only once its consequences had piled up, these problems would most plausibly have been caught and corrected while still cheap to fix.

The third is that generating code which is technically valid is not the same as generating code that makes good decisions for its context. The clearest instance of this is the password-hashing performance problem: the initial implementation used Werkzeug’s default hashing function, which is deliberately expensive to compute as a security measure, and this default is a reasonable choice in general. But applied without regard to the local development server’s single-threaded, synchronous request handling, it introduced a delay noticeable

enough to be flagged during testing. The fix — switching to bcrypt, where the hashing cost can be tuned explicitly — did not require abandoning the underlying security principle, only adjusting it to fit the actual environment the code was running in. The same pattern, in a different form, runs through the architectural problems already discussed: a service layer built from scratch for each component, a layout component that kept absorbing new responsibilities, and authentication state read independently by whichever component needed it were all technically functional at the moment each was written, and none of them represented, in isolation, a wrong decision. What they lacked was any consideration of whether the choice would still hold up as the application grew around it — which is a different kind of failure than producing code that simply does not work.

4.8 Critical Problems and the Test Failure Analysis

Principle

Beyond the problems already discussed, Part 1's codebase carried a number of further architectural weaknesses by the time it was handed over to be continued. Some of the more significant ones are summarized below.

Location	Weakness
<code>AuthContext.js</code>	User identity stored in local storage, which is readable by any script running on the page
<code>app.py</code>	Database connection string hardcoded directly in source rather than read from configuration
<code>app.py</code>	CORS configured to accept requests from any origin
<code>api_client_service.py</code>	A service function returns a framework-level HTTP response object instead of plain data, mixing response formatting into a layer that should only handle logic
<code>collection_model.py</code>	A foreign key linking a collection to its workspace was left optional when it should always be required
<code>collection_service.py</code> , <code>request_service.py</code>	Use of a deprecated ORM lookup method

One further issue — a security vulnerability in how the backend identified which user a request belonged to — is significant enough to describe in full.

The implementation built during Part 1 expected the client to pass a plain-text `user_id`, either as a query parameter (`GET /api/workspaces?user_id=1`) or within a JSON request body, and the server trusted whatever value it was given without any further verification. This created several concrete problems at once: since nothing cryptographically tied the `user_id` to the authenticated user, a request could simply change the value in the URL — for instance from `user_id=1` to `user_id=2` — and potentially view, create, or delete another user's workspaces and collections; passing an identifier this way also meant it could end up stored in browser history, server access logs, and proxy logs, none of which are places a user identifier should persist; and on the frontend, every component that needed to make an authenticated call had to fetch the user ID from storage and pass it explicitly to each service function, producing repetitive boilerplate throughout the codebase. This pattern was not an isolated oversight in one endpoint — it was how the application identified its user everywhere, and it remained the case throughout Part 1. It was left unresolved for the whole of this stage of development, and was only corrected later, when the identification mechanism was replaced with a token verified by a cryptographic signature rather than a value the client could simply supply on its own.

The second issue began as an ordinary test failure. On the first run of the collections test suite, two tests failed: one checking that all collections in a workspace were returned, and one checking that a deleted collection no longer appeared in that list. Both tests operated on the default workspace created at signup and assumed a default collection already existed inside it — but neither had ever called the collections list endpoint before performing its write operations. The actual cause was that signing up created a workspace directly, without going through the same function used to create a workspace elsewhere in the application, and that function was the only place responsible for seeding a default collection. The collection was therefore never created at signup at all; it only came into existence the first time the collections list endpoint happened to be called, since that endpoint seeded it lazily on demand.

The fix initially applied was to add a call to that same list endpoint at the start of each failing test, before its write operations ran, which reliably triggered the lazy seeding and made both tests pass. But this addressed the symptom rather than the cause: the real problem was that two different paths for creating a workspace existed in the application — signing up, and calling the workspace-creation endpoint directly — and only one of them seeded a default collection. The test had correctly caught a genuine inconsistency in the implementation; adding a setup step to the test made it pass without resolving that inconsistency, leaving the same divergent behavior in place for any other code path that relied on a workspace already having its default collection. The change that would have actually addressed the cause was making the signup path call the same workspace-creation

function used elsewhere, so that a workspace was seeded identically no matter how it was created. Listings 4.7 and 4.8 show the relevant code before and after this fix.

```
default_workspace = Workspace(name=f"{first_name}'s Space",
                               user_id=user.id, is_default=True)
db.session.add(default_workspace)
db.session.commit()
```

Listing 4.7: Signup path creating a workspace directly, bypassing `create_workspace()`.

```
# auth_service.py --- signup now delegates to the shared
workspace-creation path
create_workspace(user.id, f"{first_name}'s Space", is_default=
True)

# workspace_service.py --- create_workspace extended to always
seed a default collection
def create_workspace(user_id, name, is_default=False):
    workspace = Workspace(name=name, user_id=user_id,
is_default=is_default)
    db.session.add(workspace)
    db.session.flush() # obtain workspace.id before seeding
the default collection
    ensure_default_collection(workspace.id) # creates
collection + default requests and commits
```

Listing 4.8: Fix — signup delegates to `create_workspace()`, which is extended to always seed a default collection.

This case is what motivated the Test Failure Analysis Principle: a failing test only shows that the behavior the test expects and the behavior the implementation produces disagree, and it does not by itself say which of the two is wrong. Determining that requires comparing both against the actual system requirement, and three outcomes are possible. If the test reflects the requirement correctly and the implementation diverges from it, the implementation should be changed. If the implementation already satisfies the requirement and the test expects something that was never actually specified, the test should be changed instead. And if neither the test nor the implementation can be clearly judged against a requirement that was never precisely defined in the first place, the requirement itself needs to be clarified before either side is touched (Figure 4.5).

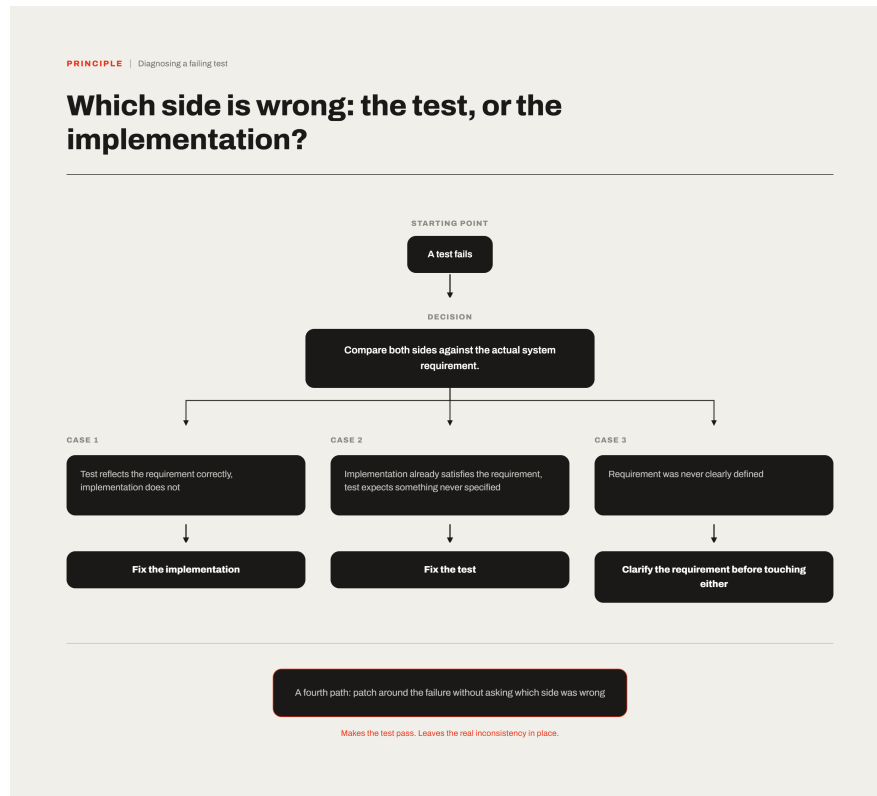


Figure 4.5: Decision flow for diagnosing a failing test: which side — the test or the implementation — holds the incorrect assumption.

5 Part 2: Structured Approach (AntiGravity)

5.1 Motivation for a Structured Approach

Part 1's architectural problems shared a common origin, and the conclusion drawn from them shaped everything that followed. The missing service layer, the `MainPage.js` monolith, the scattered reads of authentication state, the unauthenticated `user_id` parameter, and the test that was made to pass without its underlying inconsistency being addressed were not, individually, failures of the agent's capability — as Section 4.6 established, each decision was reasonable for the agent at the time it was made. What each case lacked was the same missing ingredient: a persistent point of reference to check the decision against — an existing convention to extend, a plan to follow, a review step that had to happen before the code was accepted rather than only once a problem had already spread. The conclusion drawn from this was explicit: the quality of the software produced in Part 1 had been shaped less by the coding model itself than by the architectural constraints, review process, and development rules imposed on the agent — or, in Part 1's case, by their absence.

This reframed the question for the second part of the project. Rather than treating better output as something to get by giving the agent more capability or more detailed individual prompts, Part 2 was designed around the idea that the missing ingredient was structure: a defined set of rules the agent had to follow regardless of the feature it was working on, a point at which a plan had to be reviewed before any code was written, a role dedicated to checking the result afterward rather than relying on a problem being noticed by chance, and an explicit requirement to determine why a test had failed before changing anything, rather than defaulting to whichever fix made the test pass.

Realizing this structure required an environment built to support it, which motivated the move to Google Antigravity described in Section 3.1. What this made possible — an AI Development Charter defining architectural rules before implementation began, a division of labor between an agent responsible for planning and one responsible for implementing, a mandatory design review gate before coding, a dedicated review role after it, and a formalized testing discipline — is the subject of the sections that follow.

5.2 AI Development Charter and Architectural Rules

The first concrete step taken in Part 2 was creating an AI Development Charter, `AGENTS.md` — a written set of architectural rules established before any feature implementation began, intended to be read by every agent involved in the workflow, rather than accumulated piecemeal as problems came up during development. Where Part 1's system prompt had started with only a partial set of conventions and grown reactively over the course of the project, this document existed in a mature form from the outset and applied uniformly across the whole of Part 2, not to one agent working alone but to all three roles in the workflow at once.

The rules themselves span both frontend and backend concerns, organized into named categories drawn directly from the file's own structure. On the backend: an Application Factory Pattern, Blueprint-per-Domain routing, with exactly one blueprint file per resource domain, and a Service Layer rule requiring route handlers to act only as thin coordinators while business logic and a fixed return pattern live in the service functions. On the frontend: a Service Layer rule requiring all API calls to go through `src/services/` rather than being made directly inside components, a State Ownership rule requiring shared state to be handled through Context or hooks rather than read independently wherever it is needed, and a Component Size Limit capping components at roughly 150 lines before they must be split. A further set of rules governs how a failing test must be diagnosed before being changed, formalizing, as a mandatory step rather than a retrospective lesson, exactly the reasoning the Test Failure Analysis Principle was built to describe after the fact. Drafting a rulebook this way benefits from AI assistance: producing a set of conventions broad and specific enough to cover an entire project by hand — even before any code exists to check it against — is itself a substantial task, and an agent can usefully draft a first set of conventions from established software-engineering practice as a starting point. The resulting draft was reviewed and adjusted before being adopted as binding. Listing 5.1 shows a representative excerpt of the file itself.

```
## 7. BACKEND -- ARCHITECTURAL RULES

### 7.1 Application Factory Pattern
- The Flask app MUST use the `create_app()` factory in `app.py`
- Blueprint registration happens inside `create_app()`.
- No global Flask `app` object is used outside the factory.

### 7.2 Blueprint-per-Domain
- Each resource domain has exactly one blueprint file in `backend/routes/`.
- Blueprint names follow the pattern `_bp` (e.g., `auth_bp`, `workspace_bp`).
- Route URL prefixes are embedded in the route decorators (not in `register_blueprint`).
- New resource domains MUST get a new blueprint file. Do not add routes from a new domain to an existing blueprint.
```

Listing 5.1: Excerpt of AGENTS.md showing the Application Factory Pattern and Blueprint-per-Domain rules.

Some of these rules are stated not as abstract principles but as direct references to a specific mistake already made in Part 1 — the rule requiring foreign keys to parent entities to be non-nullable, for instance, names the nullable `Collection.workspace_id` foreign key from Part 1 directly as the known violation not to repeat. The charter was, in this sense, partly written out of Part 1's own documented weaknesses rather than derived independently.

5.3 Sequential Multi-Agent Workflow (Architect / Implementer / Post-Implementation Review Agent)

The workflow adopted in Part 2 separates planning, implementing, and reviewing into three distinct roles, coordinated through a strict sequence of hand-offs rather than concurrent execution, with a mandatory human gate between two of them. This is a different axis of change from Part 1, where a single agent carried out all three activities together, in one continuous exchange, rather than any one of them being delegated to a specialized role.

The Architect Agent goes first. Its job is to analyze a feature request against the existing codebase and produce a structured `implementation_plan.md`, but it is explicitly barred from writing any implementation code itself — its output is limited to architectural plans,

interface definitions, and file-structure proposals. Listing 5.2 shows how this role's behavior is defined directly in `AGENTS.md`.

```
## 1. ROLE OF THE ARCHITECT AGENT

When acting as the Architect Agent, the agent MUST:
- Analyze the feature request against the existing codebase
  before proposing anything.
- Produce a structured implementation_plan.md artifact (see
  Section 11 for the required format).
- Include a mandatory Design Review section (Section 11.2)
  before listing any files or changes.
- NEVER write implementation code. Only produce architectural
  plans, interface definitions, and file-structure proposals.
- Flag any deviation from these rules explicitly in the plan
  and explain the justification.
- Identify which existing weaknesses (see Section 9) a new
  feature might worsen, and propose mitigations in the plan.

The Architect Agent hands off to the Implementer Agent only
after the human has reviewed the Design Review section and
explicitly approved the full plan.
```

Listing 5.2: Excerpt of `AGENTS.md` defining the Architect Agent's role and behavioral constraints.

Between the Architect and the Implementer sits a human review gate. The human is required to read the design review portion of the plan specifically; if the reasoning behind an architectural choice is vague, the human must ask for clarification, and if it is unclear who owns a given piece of state or which service is responsible for which logic, the plan must be rejected and revised rather than approved as written. Implementation cannot begin until this approval is given explicitly.

The Implementer Agent then executes, and its role is deliberately narrow: it does not design, decide, or deviate — all of that is expected to have already happened in the plan. It is restricted to creating exactly the files listed under "files to create," modifying exactly the files listed under "files to modify," and implementing exactly the endpoints and components the plan describes, even where an unlisted addition might seem like a good idea on its own. If it encounters a situation during implementation that requires a design decision the plan never addressed, it is required to stop immediately, document what was expected against what was actually found, and wait for the Architect to produce an updated plan rather than resolving the gap itself. It is also bound by every rule in the charter regardless of whether the current plan happens to mention them, required to implement in a fixed dependency order — backend models before services, services before routes, and so on through to frontend

components — and required to write the corresponding tests for anything it adds before considering the task finished. If it notices a bug or weakness while working that falls outside the current plan's scope, it is required to document it rather than fix it on the spot, since an unplanned fix is treated as its own task requiring its own plan.

Once the Implementer has finished and produced a `walkthrough.md`, the Post-Implementation Review Agent is activated. Its scope is deliberately narrow as well: it reads only the approved plan, the walkthrough, and the specific files the walkthrough lists as changed, and it does not re-check anything already enforced by the charter's rules, since the Implementer is already bound by those. Its purpose is to catch problems that could only emerge once code actually existed — the same six-point checklist already described as part of the architectural conformance criterion, covering things like a component quietly taking on responsibilities beyond what was planned, unplanned coupling between modules meant to stay independent, and code paths left untested. It produces a `review_report.md` rating each of the six concerns and closing with an overall verdict, after which a human decides whether to fix the issue immediately, defer it, or accept the feature as complete.

The first feature run through this full sequence was, fittingly, a fix to one of Part 1's own documented problems — the `MainPage.js` monolith — used as the initial test of whether the new process actually functioned as designed before being applied to newly added features.

5.4 Design Reviews Before Coding (`implementation_plan.md`)

This looks at what the plan document itself is actually required to contain, and in particular at the Design Review portion, which is the part of the document the human review gate is centered on.

An `implementation_plan.md` follows a fixed eight-part structure: a feature summary, the mandatory Design Review, a list of files to create, a list of files to modify, any API contract changes, the corresponding OpenAPI specification additions, an explicit account of any deviation from the charter's rules, and a verification plan describing how to confirm the feature actually works. Of these, only the Design Review is treated as non-negotiable — a plan missing it, or missing any of its five required sub-sections, is considered invalid and cannot be approved.

Those five sub-sections are: an architecture rationale, explaining why a particular file or component structure was chosen over the alternatives and which existing pattern it follows; a statement of state ownership, identifying, for every new piece of state the feature introduces, which component or hook owns it and why; a statement of service ownership, identifying which service file is responsible for which piece of logic and justifying whether an existing service is being extended or a new one created; a testing strategy, specifying what will be tested, at what level, and which failure scenarios are explicitly covered; and a set of scalability concerns, naming any decision that could become a bottleneck or a maintenance burden as the feature or its data grows, along with a proposed mitigation. The purpose of requiring all five before any code is written is to force the reasoning behind a design to be made explicit up front, rather than letting a plan consist of little more than a list of files to touch.

A concrete instance of this is the plan produced for adding a super-admin role to the application. Before any code existed, the plan flagged, under a dedicated review section, that the feature would require two new columns on the users table, two new database tables for audit logging and user notifications, a routing guard restricting the new admin area to admin accounts, and — notably — a privacy concern: since a super admin would be able to browse other users' saved requests and execution histories, the plan proposed a visible warning banner in that view and required every instance of an admin viewing another user's data to be recorded immutably in the audit log. In its Design Review, the plan gave each of the five required sub-sections a concrete answer for this feature: the architecture rationale justified isolating the new administrative logic under its own backend blueprint and service module rather than extending an existing one; state ownership was laid out in a table assigning each new piece of state — the admin flag itself, the active tab in the admin panel, the users and workspaces lists, in-app notifications — to the specific part of the application responsible for it; and service ownership separated the backend service handling admin actions from the frontend service wrapping the corresponding API calls.

What this example shows is what a mandatory design review actually buys in practice: a privacy concern with legal and ethical weight was identified and a mitigation proposed while the feature still existed only as a document, reviewable and rejectable by a human before a single line of code had been written — rather than being discovered only after the fact, in the way several of Part 1's problems were.

5.5 Testing Strategy: Force Requirement Validation & Three-Tier Testing

Testing in Part 2 is governed by two things: a fixed structure for what gets tested and at what level, and a mandatory reasoning step that has to happen before a failing test is touched.

The structure is a three-tier testing strategy, where each tier targets a different class of problem and none substitutes for another. The first tier is backend integration testing with `pytest`, required for every new endpoint, using the Flask test client against an in-memory database rather than the production one. The second tier is a frontend smoke test using React Testing Library, required for every new component; the requirement is deliberately minimal — asserting only that the component renders without crashing when given its expected props — kept intentionally cheap so that its cost is never a reason to skip it. The third tier is browser-based end-to-end verification, required for any feature that touches the user interface and exempted only for backend-only changes: the backend and frontend are started, the feature is exercised in the browser exactly as a user would, and the result — correct routing, UI state, error messages, and API responses — is confirmed visually, with a screenshot taken for each verified behavior and attached to the implementation's walkthrough document.

Layered on top of this structure is a rule about where tests are allowed to come from. Tests must be derived from the requirements stated in the feature's plan, not from whatever the implementation happens to do — for every requirement listed, at least one test has to verify it directly, and each requirement's coverage has to include a normal case, an error or unauthorized case, and any boundary condition the feature explicitly calls for. A feature cannot be marked complete if any requirement lacks a corresponding test, even if every test that does exist is passing. This addresses directly what had been observed as a recurring risk: that an agent, left to its own judgment, could satisfy the letter of "the tests pass" while never actually testing whether the feature does what it was asked to do. Listing 5.3 shows how this rule is stated directly in `AGENTS.md`.

```
### 8.4 Requirement-Driven Test Coverage (CRITICAL)
- Tests MUST be derived from the feature's requirements, not
  from the implementation.
- For every requirement bullet listed in the
  implementation_plan.md feature summary, the Implementer MUST
  write at least one test that directly verifies it.
- Each test suite MUST cover three scenario types:
  - Happy path: the requirement works correctly under normal
    conditions.
  - Error path: the system responds correctly when the input
    is invalid or unauthorized.
  - Boundary condition: any edge case explicitly mentioned in
    the feature request.
- The Implementer MUST NOT mark a feature complete if any
  requirement bullet has no corresponding test -- even if all
  existing tests pass.
- This rule is enforced at the Implementer level only. The
  Reviewer does not re-check requirement coverage -- doing so
  would be redundant and wasteful.
```

Listing 5.3: Excerpt of AGENTS.md defining the requirement-driven test coverage rule.

The second half of the testing strategy governs what happens when a test fails. Before making any change, the agent is required to first read the test to understand what it asserts, then read the implementation to understand what it currently does, and only then decide — and state explicitly, in writing, before touching anything — whether the fault lies with the implementation, with the test's own expectation, or, following the same reasoning already established through Part 1's collection-seeding case, with a requirement that was never clearly defined in the first place. Only once that determination has been made and recorded is either the implementation or the test allowed to be changed, and never both at once as an unexamined guess. This turns what had been, in Part 1, a lesson learned from a single incident into a mandatory procedure applied to every failing test regardless of how minor it seems. Listing 5.4 shows how this protocol is stated directly in AGENTS.md.

```
### 8.2 Agent Test Failure Protocol (CRITICAL)
When a test fails, the agent MUST follow this reasoning
protocol before making any change:

1. Read the test - understand what behavior it asserts.
2. Read the implementation - understand what the code
   currently does.
3. Determine the cause: Is the implementation wrong (bug)? Or
   is the test assertion wrong (bad expectation, outdated
   contract)?
4. Document the decision - state explicitly which one is wrong
   and why, before fixing anything.
5. Only then: fix the right artifact (implementation OR test,
   not both blindly).

The agent MUST NOT blindly change implementation to match a
failing test without this analysis.
```

Listing 5.4: Excerpt of AGENTS.md defining the agent test failure protocol.

5.6 Project Knowledge Files

As a codebase grows, an agent working on it runs into two related problems. The first is that the codebase eventually becomes too large to fit entirely into the agent's active context at once, so no single prompt can simply include everything relevant. The second is a kind of overhead that exists even before the first problem becomes acute: before writing any code, the agent has to spend time running search tools, opening files, and guessing where a given piece of logic actually lives, none of which is productive work in itself.

Part 2 addressed both by maintaining a persistent set of project knowledge files, split into two distinct kinds. The first is a rule set — AGENTS.md, described earlier — which states what an agent is and is not allowed to do, independent of where anything in the codebase happens to be located. The second kind is a pair of structure maps, BACKEND_STRUCTURE.md and FRONTEND_STRUCTURE.md, which instead describe where things currently are: the actual file and folder layout, which module is responsible for which functionality, and how the pieces of the application connect to each other. Where the rule set answers "what are the rules," the maps answer "where is everything and how is it organized" — and keeping the two separate means an agent can consult whichever kind of question it actually has, rather than searching through one large, undifferentiated document for both. Listing 5.5 shows an excerpt of FRONTEND_STRUCTURE.md's directory overview.

```

## 1. Directory Tree & File Overview

The React frontend follows a structured domain-based layout
under src/:

src/
|-- App.js                # Root router, protected route wrapper
|-- index.js              # Application entry point
|-- index.css             # Global CSS styles
|-- constants.js          # Global constants (e.g., HTTP
method colors)
|
|-- contexts/
|   |-- AuthContext.js    # Global Auth Context & Hook
|
|-- hooks/
|   |-- useWorkspace.js   # Workspace selection,
validation, loading hook
|   |-- useCollections.js # Collections/Requests CRUD
state management hook
|   |-- useWorkspaceTabs.js # Workspace tab and state
cache hook
|   |-- useWorkspaceTabs.test.js # Tab management hook tests
|   |-- useHistory.js     # Custom request history
management hook
|   |-- useHistory.test.js # History hook tests
|   |-- useInvitations.js # Custom workspace invitations
hook
|
|-- services/
|   |-- api.js            # API base configuration (BASE_URL)

```

Listing 5.5: Excerpt of FRONTEND_STRUCTURE.md showing the directory tree and file overview.

Unlike the rule set, which is written once and then only amended deliberately, the structure maps are treated as living documents that are expected to fall out of date unless something keeps them current. This is built directly into the workflow rather than left to chance: producing an updated structure map is part of what the Implementer Agent is required to output alongside its code and its walkthrough for every feature, so the maps stay synchronized with the codebase as it actually is rather than reflecting the codebase as it was when they were first written.

5.7 OpenAPI Specification Automation and Avoiding Documentation Drift

Two approaches were available for keeping the application's OpenAPI specification consistent with its actual implementation. The specification could be generated automatically from the backend source code itself, using annotations or decorators placed on each route, which guarantees consistency by construction but requires maintaining that metadata on every endpoint and ties the codebase more tightly to the generation tooling. Alternatively, the specification could remain a manually written file, with an automated test added to check it against the application's actual routes and catch any mismatch during testing rather than preventing it from being written in the first place.

Part 2 adopted the second approach. The specification, `openapi.yaml`, continued to be written and maintained by hand, but a dedicated pytest module was added that inspects the Flask application's routing table at runtime and compares it against the specification file, asserting that every implemented route and HTTP method is documented and that no documented endpoint is missing from the implementation. If a route existed in the code but not in the specification, or vice versa, this test would fail, which meant a mismatch could be caught during the same testing cycle already required for every other change rather than being noticed only when someone happened to read the specification.

This kept the backend code itself free of the extra annotations an auto-generation approach would have required, while still closing the gap that documentation drift depends on: a change that touched the API's actual surface without a corresponding update to its specification would no longer pass silently, since the routes-versus-specification check ran as part of the same testing discipline already required by the rest of Part 2's rules.

5.8 Identification of Architectural Problems and Weaknesses

Two distinct mechanisms were used in Part 2 to surface problems in the application, operating at different points in the project.

The first was the inherited weaknesses catalogued at the outset, already discussed as the starting condition Part 2 took over from Part 1. Rather than being left as background knowledge, each of these was treated as a task in its own right, going through the same Architect–human–Implementer–Reviewer sequence as any newly added feature: a plan was produced explaining how the weakness would be addressed, a human approved it before

any code was written, the fix was implemented and verified, and the result was reviewed afterward. Fixing a known problem this way was not privileged over adding new functionality — it followed the identical process, which meant a fix could be rejected at the design-review stage in exactly the same way a new feature's plan could be.

The second mechanism was a dedicated security audit carried out later in Part 2, distinct from working through the inherited list, aimed at finding problems that had not already been identified rather than resolving ones that had. This was split into two stages: a systematic review of the entire application — frontend, backend, database, and API together — to identify vulnerabilities, followed by a separate stage in which each identified vulnerability was patched and the fix specifically verified rather than merely applied and assumed correct. Structuring the audit this way kept discovery and remediation as separate steps, so that a vulnerability had to be explicitly confirmed before any change was made in response to it, rather than a fix being written on the basis of a suspicion that turned out to be wrong.

Together, these two mechanisms meant that problem identification in Part 2 was not something that happened only once, at the project's outset. The initial catalogue addressed what had already been observed; the later audit was a deliberate second pass, run independently of any prior list, checking for what a first pass might have missed.

6 Comparison of Both Approaches

6.1 Comparison: Part 1 vs. Part 2

As established in Section 3.1, both parts of the practical project built the same application with the same underlying model, Claude Sonnet 4.6, held constant so that any differences between the two parts could be attributed to how the collaboration was structured rather than to a difference in model capability. Part 2 did not start over: it continued Part 1's codebase, inheriting its architecture, its problems, and its catalogued weaknesses. What changed was not the software being built or the model building it, but the process around it.

The persistent context document each part relied on differed in when its rules existed relative to the problems they addressed. CLAUDE.md began with only a partial set of conventions and grew reactively, each addition following a problem already encountered. AGENTS.md existed in a mature form from the outset and bound all three agent roles uniformly; several of its rules — the non-nullable foreign key requirement is one example — are traceable directly to a specific mistake already made in Part 1, rather than arrived at independently.

The two parts also differ in whether a design was reviewed before implementation began. Part 1 had no per-feature planning step: prompts were sequenced ad hoc, and a design was never drafted and approved ahead of writing code. Part 2 made this mandatory through `implementation_plan.md`'s Design Review, which cannot be skipped or left incomplete, and which surfaced at least one concern — the privacy implications of the super-admin role — before any code existed to review.

This is inseparable from a shift in who did the work. Part 1 was a single agent working through one continuous thread, handling requirements interpretation, implementation, and whatever review occurred together. Part 2 split these into three roles — Architect, Implementer, Post-Implementation Review Agent — coordinated through structured artifact handoff rather than one undifferentiated exchange.

Human oversight moved correspondingly. In Part 1 it existed but operated after output already existed, and in practice changes were accepted once they worked rather than checked

against the system they joined; oversight only became substantive once a problem was already obvious, as with `MainPage.js`'s growth. In Part 2, human approval is a gate before implementation begins, and a vague or incomplete plan must be rejected rather than corrected later.

Testing followed the same trajectory from informal to formalized. Part 1's tests were backend-only, written after the implementation they verified, and evaluated once, after the fact, along five ad hoc dimensions; no frontend tests existed at all. Part 2 replaced this with a fixed three-tier strategy and a rule tying every test to a stated requirement rather than to whatever the implementation happens to do, so passing tests alone no longer imply a requirement was actually covered.

The handling of a failing test shows the same pattern at a smaller scale. Part 1's Test Failure Analysis Principle emerged from one specific incident — a collections test that was made to pass by patching around a workspace-seeding inconsistency rather than fixing it — and was applied after the fact, as a lesson drawn from that case rather than a standing rule enforced elsewhere in the project. Part 2 wrote the identical reasoning into the Charter as a mandatory step: before changing anything, an agent must read the test, read the implementation, and explicitly determine and record whether the fault lies with the implementation, the test's own expectation, or an underlying requirement that was never clearly defined — required for any failing test, regardless of how minor the failure looks, rather than reserved for cases serious enough to prompt that reasoning on their own.

Accumulated weaknesses were also handled differently. In Part 1, each major architectural problem was only identified once it had already spread, and fixed as a retrofit; fifteen further weaknesses were catalogued and left unresolved at handoff. In Part 2, addressing an inherited weakness went through the same Architect–human–Implementer–Reviewer sequence as any new feature — fittingly, the first feature run through that sequence was a fix to the `MainPage.js` monolith itself — and a separate, independent security audit added a second detection mechanism beyond the inherited list.

Finally, documentation consistency: Part 1's OpenAPI specification had no automated check tying it to the implementation. Part 2 added a `pytest` module comparing the specification against the application's actual routes at runtime, folding drift detection into the same testing cycle already required for everything else.

Across all of these, the pattern is consistent: what could be built did not change, since the same model produced both parts, but where a decision was checked, by whom, and at what point in the process shifted from informal and retrospective to formalized and positioned ahead of implementation. Whether that shift produced better software, rather than more process, is taken up in the evaluation that follows.

6.2 Feature Diversity

Part 1's feature set and Part 2's additions differ not just in what each does, but in what kind of engineering problem each one poses to whoever is building it. That distinction is the subject of this section.

Part 1's feature set is structurally uniform beneath its range:

- Authentication, workspace management, collection management, and request management are each a create/rename/delete operation on a resource the current user owns.
- The request builder is a configuration form: a method and URL, key-value tables for parameters and headers that auto-add a row as the last one fills, a body tab offering none, form-data, x-www-form-urlencoded, or raw text, and an authorization tab injecting a header for no auth, a bearer token, or basic auth.
- Request execution fires that configuration as a single HTTP call.
- The response panel formats whatever comes back — status code, timing, pretty-printed JSON.
- Sidebar search filters, client-side, across records the user already owns.

None of this requires reasoning about another user's data, aggregating many records into a derived value, or reacting to events elsewhere in the system — every feature reduces to "manage one resource" or "fill in and submit a form."

Part 2's additions each introduce a shape Part 1 never had to handle:

- Roles — Super Admin globally, Owner/Editor/Viewer at the workspace level — are access-control logic that branches on who is acting relative to whose data, not on whether a record belongs to the current user, and this logic has to be threaded through nearly every existing endpoint and component rather than living in one new file.
- Inline request comments introduce threaded, collaborative content shared by multiple users on the same request at once, a data shape and a concurrency concern present in no Part 1 feature.
- Activity feed and logging is a cross-cutting observability feature by construction: a single chronological ledger has to receive events from settings changes, collection and request edits, membership changes, and execution runs alike, so the feature's

correctness depends on every one of those other subsystems correctly emitting an event, not on any logic contained within the feature itself.

- Performance and analytics is an aggregation feature — KPIs, status-code distributions, and per-endpoint analysis computed over a window of past execution records, rather than a read or write of a single one.
- Request export is a stateless, client-side data-transformation utility: it serializes the same request-configuration state the builder already holds into a cURL string or a downloadable JSON file, with no backend persistence involved at all.
- OpenAPI automation is not a user-facing feature in the first place — it is tooling that checks two other artifacts, the routing table and the specification file, against each other.

Part 1's feature set reduces to one recurring shape: a single resource, owned by one user, managed through create, rename, and delete, and displayed through a configuration form. Part 2's additions moved well beyond that shape — access control based on who is acting rather than who owns the data, content shared live between multiple users, a ledger fed by every other part of the application, aggregation over a history of past records, a stateless serializer with no backend state, and a consistency check between two non-code artifacts. Each of these represents a genuinely different kind of task, not a variation on the one Part 1 had already solved.

7 Evaluation

7.1 Evaluation Criteria

Four criteria were already defined before either part of the project began: specification satisfaction, test validity, architectural conformance, and process and effort. Each is assessed here the same way it was already defined — specification satisfaction by checking behavior against the original requirement rather than against whichever tests happen to exist, test validity through the same three-way check used earlier, architectural conformance against the six underlying concerns the checklist formalizes, checked fully and by rule in Part 2 and only partially and informally in Part 1, and process and effort against the milestone and task records kept throughout both parts.

This chapter applies those four criteria one at a time, using the material already established for both parts rather than introducing new examples. Each criterion gets its own verdict for Part 1 and its own verdict for Part 2, and those verdicts are brought together at the end into a single summary assessment.

7.2 Quality of Generated Code

This section applies specification satisfaction — whether a feature’s behavior fulfills its functional and non-functional requirements and handles its edge cases, not merely whether its tests pass — to Part 1 and then to Part 2.

Part 1’s functional requirements were generally met, but rarely on the first attempt and rarely without a gap the prompt itself had left open. The missing email field, the missing uniqueness constraint on it, and the undifferentiated validation messages in the login and registration feature were each caught only once the implementation was reviewed against what the feature was actually meant to do, not against what had literally been asked for — meaning specification satisfaction in Part 1 depended on a human noticing the gap after the fact, feature by feature, rather than on anything in the process that would have surfaced it beforehand. Non-functional requirements fared worse. The unauthenticated `user_id` parameter was a direct failure of the security requirement — any user could view or modify

another user's data simply by changing a number in the URL — and unlike the login-feature gaps, which were caught and fixed once someone reviewed the output, this one was never caught during Part 1 at all. The three architectural weaknesses — the missing service layer, the MainPage.js monolith, and the scattered authentication reads — represent the same failure at the level of maintainability: each satisfied its immediate functional requirement while working against the maintainability the non-functional requirements called for, and none of the three was caught until its consequences had already spread across multiple files. Part 1's code, in short, tended to satisfy what a feature was explicitly asked to do and to fail, sometimes for the whole duration of the phase, at the requirements that were assumed rather than stated.

Part 2 shifted where that gap could still occur. The mandatory Design Review moved the check earlier — the super-admin plan's privacy concern was identified and mitigated while the feature existed only as a document, not after code already existed to review — and several Charter rules encode a specific Part 1 miss directly. Part 2's added scope was also structurally more varied than Part 1's, meaning specification satisfaction had to hold across a wider range of requirement types rather than repeated instances of one. Part 2's code, on the whole, satisfied both its functional and its non-functional requirements more consistently, and — where the Design Review gate worked as intended — pre-emptively, rather than only once a gap had already become visible in the running application.

In short, both parts were held to the same standard, but Part 1 met it after the fact, while Part 2 met it — at least for its highest-stakes cases — in advance.

7.3 Quality and Validity of Generated Tests

This section applies test validity — whether a test suite actually verifies the intended behavior, not just whether it runs and passes — to Part 1 and then to Part 2.

Part 1's backend testing was not limited to a single feature; pytest suites were generated after implementation across the application's routes, services, and models more broadly. The five evaluation dimensions used in this thesis — coverage, correctness, completeness, an overgeneration check, and debugging capability — were applied informally to these suites as well. The collections test-failure case (Section 4.8) sits outside these five dimensions entirely: the test itself was correct, but the fix that followed patched around the failure instead of resolving it, which is the distinct problem test validity was introduced to address.

Part 2's three-tier structure requires a pytest integration test for every new endpoint, a smoke test for every new component, and browser-based verification for anything touching

the interface, so coverage no longer depended on whether an agent happened to generate tests for a given part of a feature. The requirement-derived testing rule targets validity rather than coverage: a test suite is no longer complete because it runs and passes, but only once every requirement bullet in the feature's plan has a corresponding test, covering a normal case, an error case, and any stated boundary condition — a test checking something true but irrelevant to the requirement no longer counts. The permission-denial verification captured for the workspace roles feature — a viewer blocked from creating a collection, a non-owner blocked from inviting a member — is a validity concern specific to access-control logic that nothing in Part 1's feature set ever required testing for. And the reasoning behind the collections case, informal and applied once in Part 1, became a mandatory step in Part 2, required for every failing test regardless of how minor it looks.

Across both parts, test validity moved from something assessed informally, feature by feature, to a structural requirement checked the same way for every feature — and from a suite that could pass while checking too little or checking the wrong thing, to one where a passing result is tied explicitly back to a stated requirement rather than to whatever the implementation happens to do.

7.4 Architectural Conformance

This section applies architectural conformance against a fixed six-point checklist, used consistently for both parts regardless of when it was formally introduced. Part 1's process checked only two of the six points — single-responsibility violations and unintended coupling between modules — and did so occasionally, as problems surfaced, rather than by rule. Part 2 extended checking to the remaining four points — untested code paths, excessive prop-drilling, unused or dead state, and inconsistent naming — and applied all six by rule, for every feature, through its Post-Implementation Review Agent.

Part 1's two documented cases show what "occasionally, as problems surfaced" meant in practice. The MainPage.js monolith is a direct single-responsibility violation: at 186 lines it already owned three distinct regions — the top bar, the sidebar, and the main panel — before the remaining planned features had even been implemented. The missing service layer and the scattered authentication-state reads are both instances of unintended coupling: components were tightly bound to their own fetch logic and their own direct reads of local storage, rather than depending on one shared place for either concern.

Part 2 checked all six categories for every feature, by rule, through the Post-Implementation Review Agent's review_report.md, which rates each of the six concerns individually and closes with an overall verdict before a human decides whether to fix, defer, or accept. The

four categories Part 1 never checked — untested code paths, excessive prop-drilling, unused or dead state, and inconsistent naming — only became part of architectural conformance with Part 2’s checklist. The review recorded for the workspace-level roles feature shows all six points checked together for one feature, reproduced in Figure 7.1.

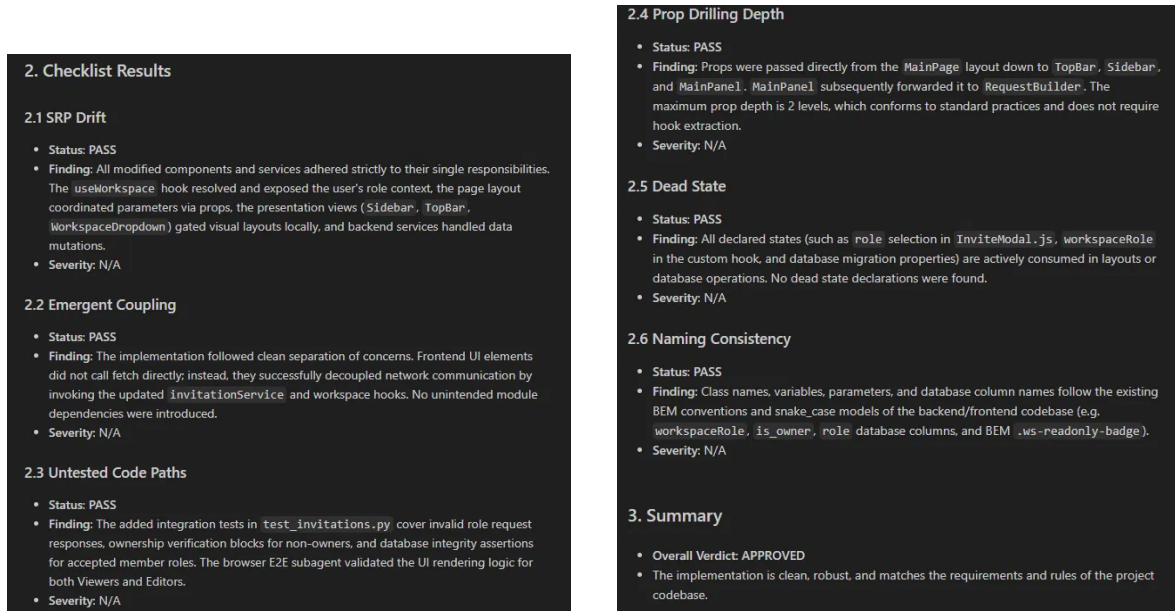


Figure 7.1: Post-Implementation Review Agent’s six-point architectural checklist for the workspace-level roles feature, recorded in `review_report.md`.

Taken together, Part 1’s conformance checking covered two of the six categories, both caught only once already widespread. Part 2’s conformance checking covers all six, by rule, and the workspace-roles review shows it actually happening in full for one feature, not only guaranteed in principle.

7.5 Efficiency and Effort (Human vs. Agent)

This section applies process and effort to Part 1 and then to Part 2 — not just how much correction a feature needed, but who did the work of catching the problem in the first place.

In Part 1, the human carried that burden almost entirely. The three architectural retrofits were each only caught because a person noticed the pattern spreading and added a rule to `CLAUDE.md` — the agent itself never flagged the missing service layer, the growing `MainPage.js`, or the scattered `localStorage` reads. The collections test case is the same pattern at a smaller scale: the agent produced a fix that made the test pass, and it took a human reasoning through the failure afterward to find what the fix had actually missed.

The agent's role throughout was executing a correction once told what was wrong, not identifying that something was wrong in the first place.

Part 2 shifted a share of that detection work onto the agents themselves. The super-admin plan's privacy concern was not caught by a human reviewing finished code — it was flagged by the Architect Agent, inside the plan, before any code existed; the human's role there was to review and approve a concern already surfaced, not to discover it. The Post-Implementation Review Agent's six-point checklist works the same way: it inspects the modified files and produces a rated report on its own, and the human's job narrows to deciding whether to fix, defer, or accept what the report already found.

The two parts differ less in total effort than in its division. Part 1's human did the diagnosing and the agent did the fixing once told. Part 2's agents did more of the diagnosing, and the human's role moved from catching problems to approving what the agents had already caught.

7.6 Summary Assessment: Part 1 vs. Part 2

Chapter 7 applied four criteria to both parts. The table below brings the four verdicts together.

Criterion	Part 1	Part 2
Specification Satisfaction	Satisfied what was explicitly asked; missed what was assumed, sometimes for the whole phase.	Caught more gaps earlier; treated inherited gaps as work to complete rather than debt to carry.
Test Validity	Assessed informally, feature by feature; a passing suite could still check too little or the wrong thing.	A structural requirement for every feature; a passing result tied explicitly to a stated requirement.
Architectural Conformance	Checked two of six categories, only once a problem had already spread.	Checks all six by rule, demonstrated in full for at least one feature.
Efficiency and Effort (Human vs. Agent)	Human diagnosed problems; agent executed fixes once told.	Agents surfaced more problems themselves; human's role shifted to approving what was already found.

Across all four criteria, the same shift recurs: from problems caught late, informally, and mostly by a human, to problems caught earlier, by rule, with the work of detection shared more evenly between human and agent.

8 Best Practices for Human-Agent Collaboration

8.1 Importance of Prompt Precision and Context Design

The foundations chapter separated prompt precision from context design as two distinct gaps an instruction can leave open: precision concerns whether an instruction states the actual requirement clearly, while context concerns whether the agent has been given the surrounding information — conventions, existing patterns, project-specific constraints — it needs to act on that instruction consistently with the rest of the system. The practical project's two parts show what happens when each gap is left unaddressed, and what changes when it is closed deliberately.

Part 1 demonstrated the precision gap directly: output quality tracked how precisely a prompt specified what was wanted, at the level of both visual detail and missing functional requirements, and the resulting gaps were not failures of capability but direct consequences of what the prompt had left unstated. The practical implication is straightforward but easy to underweight in the moment a prompt is written: an agent does not signal that a requirement was ambiguous by producing a visibly uncertain or inconsistent result. It settles on one interpretation and produces output that looks complete and correct, which means the burden of precision falls entirely on whoever writes the prompt, before any output exists to check against. **A recommendation that follows directly from this is to state assumptions explicitly rather than leaving them implicit — particularly for requirements a human would consider too obvious to need spelling out, since those are exactly the ones most likely to be silently reinterpreted.**

Part 1 also illustrates the context gap, though less explicitly than the precision gap, since CLAUDE.md began with only a partial set of conventions and grew reactively rather than supplying project context up front. Part 2's response to this was architectural rather than a matter of writing better individual prompts: rather than repeating relevant context inside each task instruction, it maintained a persistent set of project knowledge files split by kind — a rulebook stating what an agent is and is not allowed to do, and structure maps stating

where things currently are and how they connect — kept synchronized with the codebase as part of the standard implementation workflow rather than left to go stale. **The best practice this points to is institutionalizing context design at the level of the project rather than the level of the individual prompt: maintain a standing, up-to-date source an agent can consult for conventions and structure, rather than relying on each task instruction to restate them correctly from scratch.**

Taken together, the two parts suggest that precision and context should be treated as separable practices rather than as one problem solved by writing more detailed prompts. A precise instruction given without adequate context still leaves an agent to invent conventions it has no way of knowing about; thorough context supplied alongside a vague instruction still leaves the actual requirement unresolved. **Addressing both — stating the requirement explicitly, and maintaining a persistent source of project context the agent can draw on — is, on the evidence of both parts, a precondition for the agent's output being checked against what was actually wanted rather than against a plausible-looking guess.**

8.2 Role of Human Oversight (Human-in-the-Loop)

The foundations chapter distinguished human-in-the-loop as a deliberate design choice — a decision about where, in an otherwise automated process, human judgment is required before the process can continue — from oversight understood loosely as "a person eventually looked at the result." It also distinguished several forms this can take: automatic acceptance with no check at all, review after a finished output already exists, and a genuine gate that pauses the process and waits for explicit approval before it proceeds. The two parts of the practical project sit at different points along this range, and the difference in where oversight was placed, not merely whether it existed, is what separated the two.

Part 1 had a standing rule requiring the agent to ask before modifying any existing file, but in practice this amounted to review after the fact rather than a gate: changes were generally accepted once they worked, without being checked against the system they were being added to, and oversight only became substantive once a problem was already obvious enough to notice on its own — by which point correcting it meant undoing an accumulation rather than preventing one decision. **The practical cost of this placement is not that oversight was absent, but that it arrived too late to be cheap: the same logic that makes a requirements-stage mistake far less costly to fix than one caught after implementation applies directly here, and review positioned after output already exists forfeits exactly that advantage.**

Part 2 moved the same kind of judgment earlier, into a genuine gate rather than a later check: a human is required to approve the Design Review portion of a plan before any implementation begins, and is explicitly instructed to reject a plan outright — not correct it after the fact — if the reasoning behind an architectural choice is vague or if it is unclear which component or service owns a given piece of state or logic. **The recommendation this supports is to place mandatory approval at the point where a decision is still cheap to reverse — before code exists to be reworked — rather than relying on review to catch the same problem once it has already been built on top of.**

The division of labor also shifted, and this shift is itself a form of oversight design worth naming as a practice. In Part 1, a human did the work of noticing that something was wrong in the first place, and the agent's role was limited to executing a correction once told. In Part 2, the Architect Agent surfaced at least one significant concern — flagged inside a plan, before any code existed — that a human then reviewed and approved rather than discovered independently, and the Post-Implementation Review Agent's report narrowed the human's role after implementation to deciding whether to fix, defer, or accept a finding the agent had already produced on its own. **This suggests that human authority is best reserved for judgment a structurally different party cannot supply — approving or rejecting a plan, weighing a trade-off, deciding what to do about a flagged problem — while the work of detecting candidate problems in the first place can be delegated to an agent role built specifically for that check, provided a human still holds the authority to act on what it finds.**

8.3 Architectural Guardrails for AI Agents

The foundations chapter distinguished architecture erosion, caused by decisions that directly violate a governing architecture, from architecture drift, caused by decisions made without any awareness that a governing architecture exists at all — and noted that this is a cumulative problem rather than a single-instance one: no individual decision needs to be wrong on its own terms for a codebase to gradually lose its coherence, since each decision may correctly satisfy the specific instruction it was given while still failing to align with a structural pattern established elsewhere. This distinction between erosion and drift matters for what kind of guardrail actually addresses the problem, because the two parts of the practical project differ precisely on whether a governing architecture was made explicit and enforceable in the first place.

Part 1's architectural problems are better characterized as drift than erosion: the missing service layer, the MainPage.js monolith, and the scattered reads of authentication state

were not violations of a stated rule, because no rule existed yet to violate — each decision was reasonable in isolation, and what was missing in every case was something persistent to check the decision against. **The guardrail this points to is not "write better individual instructions" but "establish an explicit, standing architecture before implementation begins," since drift specifically arises where no governing structure was ever articulated for a decision to be measured against.**

The Charter is Part 2's answer to this, and its rules are concrete rather than aspirational: a single application-factory pattern for the Flask application, exactly one blueprint file per resource domain, route handlers restricted to thin coordination rather than containing business logic, and a fixed return pattern for service functions rather than each one inventing its own error-handling convention. Several of these rules are traceable directly to a specific mistake already made in Part 1 — the requirement that foreign keys to parent entities be non-nullable names the nullable `Collection.workspace_id` foreign key from Part 1 as the violation not to repeat. **This is itself a best practice worth naming separately: a guardrail written as a direct response to a documented failure is more concrete and more enforceable than one derived abstractly, since it can be checked against a specific, known case rather than against a general principle open to interpretation.**

A rule's existence does not by itself guarantee conformance, and Part 2's structure addresses this by pairing the Charter with a role whose entire responsibility is checking adherence to it after code exists: the Post-Implementation Review Agent applies a fixed six-point checklist to every feature, covering categories — untested code paths, excessive prop-drilling, unused or dead state, inconsistent naming — that Part 1's process never checked at all, rather than only the two categories Part 1 caught occasionally and only once a problem had already spread. **The guardrail this establishes is that architectural rules need a standing, rule-based check paired with them to remain effective — a rule that is stated once but never systematically verified affords no more protection against drift than having no rule at all, since an agent working under time or scope pressure has no structural reason to notice a quiet departure from it.**

8.4 Recommendations for Multi-Agent Workflows

The foundations chapter described role specialization as trading the simplicity of a single continuous loop for narrower, more focused reasoning at each stage, coordinated through structured hand-off of artifacts rather than open-ended conversation — and it flagged a specific cost that comes with this trade: an error introduced early in the pipeline can silently

propagate through every later stage rather than being caught immediately, particularly if a later agent has no way of seeing the original task and can only work from what the previous agent handed it. Part 2's Architect–human–Implementer–Reviewer sequence is a concrete instance of this pattern, and the way it was structured is worth drawing recommendations from directly, since it addresses the propagation risk rather than simply accepting it.

The role boundaries themselves were kept deliberately narrow and non-overlapping: the Architect Agent is explicitly barred from writing implementation code, limited to producing plans, interface definitions, and file-structure proposals, while the Implementer is restricted to creating and modifying exactly the files the approved plan lists, and is required to stop and request an updated plan rather than resolve any gap the plan never addressed itself. **A recommendation that follows from this is to define each agent role by what it is not permitted to do as much as by what it is responsible for, since a role with authority to quietly expand beyond its planning or implementation boundary reintroduces the single-agent problem of one actor carrying every concern at once, inside what looks like a multi-agent structure.**

The propagation risk the foundations chapter describes is addressed most directly by where the human gate sits relative to the hand-off between roles, rather than by anything internal to how the agents communicate. The plan's Design Review is required to contain five specific sub-sections — architecture rationale, state ownership, service ownership, testing strategy, and scalability concerns — and a plan missing any of them is treated as invalid and cannot be approved, which means the human reviewing the hand-off from Architect to Implementer has a fixed, comparable structure to check each time rather than an open-ended document to interpret freely. **This suggests that a multi-agent workflow's hand-off points, not just its individual roles, need an explicit, enforced structure to review against; without one, a human gate can be nominally present while still allowing an incomplete or vague plan through, at exactly the point where an error would otherwise begin propagating into every stage built on top of it.**

The workflow also builds in a mechanism for what happens when an agent's scope proves too narrow for something it encounters — a case the foundations chapter's error-propagation concern does not by itself resolve. Rather than having the Implementer either overstep its boundary to fix an unplanned issue or silently ignore it, the workflow requires the issue to be documented as its own candidate task, to be picked up through the same plan-review-implement sequence rather than folded into whatever is currently in progress. **The practice this generalizes is to give every agent role an explicit escape valve for work that falls outside its defined scope — a required documentation step rather than an implicit choice between overstepping the boundary or leaving the issue**

unaddressed — since narrow roles are only safe to enforce strictly if there is a defined place for what they are not allowed to handle themselves.

8.5 Avoiding Documentation Drift

The foundations chapter identified why documentation drift is structural rather than accidental: code changes are continuously exercised by compilers, automated tests, and continuous integration pipelines, so problems introduced into the code tend to be caught quickly, while a specification file has no built-in awareness that the code it describes has changed, and is instead updated manually, opportunistically, and often incompletely. This framing matters for what kind of practice actually prevents drift, because it rules out relying on manual diligence alone — the same absence of automatic checking that allows drift to occur in the first place is exactly what manual diligence has no structural way to compensate for consistently.

Part 1's OpenAPI specification had no automated check tying it to the implementation at all, which means any divergence between what the specification claimed and what the routes actually did would only be noticed if someone happened to compare the two directly — the same failure mode the foundations chapter describes as the general cause of documentation drift, present here in its plainest form. **The practice this rules out is treating documentation accuracy as something to be maintained by discipline or convention alone; without an automatic check, a specification's correctness depends entirely on someone remembering to update it every time, which is precisely the condition under which drift accumulates unnoticed.**

Part 2 addressed this not by generating the specification automatically from the code, but by keeping the specification hand-written and adding a dedicated pytest module that inspects the Flask application's routing table at runtime and compares it against the specification file, asserting that every implemented route and method is documented and that no documented endpoint is missing from the implementation. **The recommendation this supports is to fold a consistency check between code and documentation into the same automated pipeline that already checks the code itself, rather than either generating documentation automatically or trusting it to manual updates — a mismatch then produces a failing test, using the same enforcement mechanism already relied on for correctness elsewhere, instead of depending on someone to notice a discrepancy on their own.**

This choice also reflects a trade-off worth stating explicitly rather than treating the decision as automatic. Generating a specification directly from source-code annotations guarantees

consistency by construction, but requires maintaining that metadata on every endpoint and ties the codebase more tightly to the generation tooling; keeping the specification as a separate, hand-written artifact avoids that coupling, at the cost of needing an explicit check to catch what automatic generation would have prevented from arising at all. **The practice this points to is not that one approach is unconditionally correct, but that whichever approach is chosen, the choice should be paired with a deliberate mechanism — either generation-by-construction or an automated comparison — rather than left as an assumption that a document and the code it describes will simply stay in agreement on their own.**

8.6 Cost and Token Efficiency in Agent-Based Workflows

An agent's cost depends more on how much context builds up over a task than on any single request. Each step in an agent's work resends the conversation so far, the tool definitions it has access to, and any project context it was given, so the longer or more complex a task gets, the more expensive every additional step becomes — and generated output typically costs several times more per token than the input, so unnecessarily long or repetitive agent output adds up faster than it might seem.

A first practice is reusing stable, repeated context instead of resending it in full every time. Part 2's Charter and structure maps are a concrete instance of this: instead of retyping the project's architectural rules and conventions into every task prompt, they were kept as standing files that an agent read at the start of a task. This avoided resending the same rules and conventions in full each time a new feature was requested, and is usually the easiest cost saving to put in place relative to the effort it takes.

A second practice is matching each task to the model it actually needs, instead of using the most capable model for everything. Simple, well-defined subtasks can be handled by a smaller or cheaper model, while a stronger model is reserved for steps that genuinely require deeper reasoning. This mirrors the same idea behind splitting work across specialized agent roles, just applied to choosing a model instead of dividing up a task.

A third practice is being deliberate about what context is actually included in each step, rather than letting it grow by default. Part 2's Charter applies this directly to the Reviewer Agent: rather than exploring the broader codebase, it is restricted to reading exactly three named artifacts — the approved implementation plan, the walkthrough, and the specific files listed as changed — with the Charter stating this restriction exists explicitly

to keep the review "focused and token-efficient." Pulling in only the specific information a step needs, instead of full documents or the entire codebase every time, keeps cost from creeping up simply because it was easier not to trim it.

9 Discussion

9.1 Opportunities and Limitations of AI Agents in Software Development

Across both parts, the project shows that AI coding agents can do substantial, genuinely useful work. They produced complete features spanning backend and frontend, generated systematic test coverage for non-trivial scenarios, and — once given a fixed, narrow checklist instead of an open-ended instruction to "review the code" — reliably found problems a human reviewing informally had missed, such as untested code paths or inconsistent naming. This points to a real opportunity: an agent cannot replace the judgment a human reviewer brings, but a large share of the mechanical, checklist-style part of quality review can be handed to an agent built specifically for that job, leaving human attention for the judgment calls a checklist cannot make on its own.

Part 1 had almost no structure in place, so it is tempting to assume that the moments where a human had to step in and exercise judgment were simply a consequence of that missing structure — and that adding a Charter and a review process, as Part 2 did, would make that need for judgment go away. The project's evidence says otherwise. Even with the Charter and the mandatory Design Review in place, a human was still needed to approve or reject a plan's reasoning, or to decide what a vague ownership boundary should resolve to. That need for judgment did not disappear under a more structured process; it simply moved earlier and became concentrated at fewer decision points, mainly at plan approval, instead of being scattered throughout implementation the way it was in Part 1. So the claim this supports is a narrower one: structure changes when and how human judgment gets used in an AI-assisted workflow, not that it removes the need for that judgment.

A second limitation concerns where the Charter's rules actually came from. Several of its concrete rules — the non-nullable foreign key requirement, the service-function error pattern, the component size limit — trace directly back to specific mistakes already made in Part 1. This matters for how far Part 2's success can be generalized: the structured approach worked well partly because its rules were written with firsthand knowledge of

exactly what had gone wrong in this codebase, using Part 1 as the source of the concrete failure modes each rule was designed to prevent.

9.2 Implications for Practice

The clearest practical implication is that adopting a more structured, multi-agent workflow is not primarily a tooling decision. Google Antigravity supported parallel, role-specialized agents, but the actual source of Part 2's improvement was the Charter and the review structure built on top of that support — a set of project-specific rules and gates that had to be authored, and that did the substantive work regardless of which environment executed them. An organization considering a similar move should expect to invest in writing and maintaining this kind of rulebook as a first-class deliverable, not treat it as a byproduct of switching to a more capable agent platform.

A second implication follows from where the Charter's rules came from. Not all of it needs to be learned from scratch: a rulebook can reasonably start from architectural conventions that are already widely accepted — established patterns like a single application factory, thin route handlers, or a consistent service-layer error format are not specific to this project and could be adopted upfront from general software-engineering practice or from rules carried over from earlier projects. What this project adds on top of that starting point is the second source the Charter actually drew on: rules written directly from mistakes observed during Part 1, such as the non-nullable foreign key requirement. For a new project, this suggests treating a rulebook as built in two layers — established conventions adopted from the outset, and project-specific rules added as concrete failure modes are discovered during development — rather than either writing everything in advance or waiting to learn everything the hard way.

A third implication concerns where oversight is placed rather than how much of it exists. The comparison in this thesis indicates that moving a human's approval earlier — to a plan, before implementation, rather than to a finished result — was more consequential than any change in how much the human reviewed. For practice, this reframes human-in-the-loop design as a question of sequencing: deciding at which stage of an agent's workflow a human is required to act, rather than only deciding whether review happens at all.

10 Conclusion and Outlook

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